

Attention as Gauge-Theoretic Variational Inference

Supplementary Material

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This supplement collects derivations, implementation details, and additional empirical analyses that support the main paper but are not essential for following the core argument. Appendix A reviews standard differential geometry for readers from machine learning backgrounds. Appendix B derives the covariance gradient and its equilibrium analysis. Appendix C derives the gauge frame gradients for both $SO(N)$ and $GL(K)$, including pre-conditioning strategies. Appendix D details the numerical methods used for variational gradient descent on the Gaussian and gauge-frame manifolds. Appendix E presents the full BERT validation of the degenerate-limit predictions, including the transformer validation protocol, detailed empirical results across 105 passages and five architectures, and Bayesian uncertainty quantification. Appendix F develops the renormalization group universality conjecture and its numerical validation. Appendix G reports the symmetry breaking simulations and model-channel formalism. Appendix H proves the conditional uniqueness of the forward KL divergence via variational duality.

Appendix A. General Mathematical Framework

A.1 Principal Bundle and Associated Bundles

The following geometric and probabilistic constructions comprise standard methods in differential geometry and gauge theory. See references for details, proofs, and examples (Nakahara, 2003; Frankel, 2011; Blei et al., 2017a; Amari, 2016).

Let $\pi : \mathcal{N} \rightarrow \mathcal{C}$ be a smooth principal G -bundle where $\mathcal{C}$ is a smooth manifold (the base space) and G is a Lie group (the structure group) acting freely and transitively on the right on $\mathcal{N}$. The projection satisfies $\pi(n \cdot g) = \pi(n)$ for all $g \in G$, $n \in \mathcal{N}$.

Bundle triviality. Throughout the main paper we work with the vertex-frame parameterization $\Omega_{ij} = \exp(\phi_i) \exp(-\phi_j)$, which encodes a *globally trivial* principal G -bundle: a single chart $\mathcal{N} \cong \mathcal{C} \times G$ suffices up to gauge equivalence, and the curvature 2-form of the induced connection vanishes identically (Lemma ?? of the main text). The principal-bundle vocabulary developed in this appendix is therefore strictly more general than the parameterization used by the language model and the BERT validation, both of which live in the flat-bundle subclass. We retain the bundle apparatus because the edge-relaxed extension $\Omega_{ij} = \exp(\phi_i) \exp(\delta_{ij} \cdot G) \exp(-\phi_j)$ promotes the bundle to a non-trivial principal G -bundle in which the cocycle condition can fail and a Wilson observable on closed loops is non-trivial; this is the Regime II discussed in the main text.

Let $\rho_q : G \rightarrow \text{Aut}(\mathcal{B}_q)$ and $\rho_p : G \rightarrow \text{Aut}(\mathcal{B}_p)$ be representations of G on smooth statistical manifolds $\mathcal{B}_q$ (belief/recognition fiber) and $\mathcal{B}_p$ (model/prior fiber). These fibers are

typically K -dimensional probability simplices Δ^K (for categorical distributions) or statistical manifolds with information-geometric structure (e.g., Gaussian manifolds, exponential families).

The associated bundles are:

$$\mathcal{E}_q := \mathcal{N} \times_{\rho_q} \mathcal{B}_q = (\mathcal{N} \times \mathcal{B}_q) / \sim_q, \quad (1)$$

$$\mathcal{E}_p := \mathcal{N} \times_{\rho_p} \mathcal{B}_p = (\mathcal{N} \times \mathcal{B}_p) / \sim_p, \quad (2)$$

where $(n \cdot g, b) \sim_u (n, \rho_u(g)b)$ for $u \in \{q, p\}$.

These give fiber bundles $\pi_{\mathcal{E}_q} : \mathcal{E}_q \rightarrow \mathcal{C}$ and $\pi_{\mathcal{E}_p} : \mathcal{E}_p \rightarrow \mathcal{C}$ with fibers $\mathcal{B}_q(c) \cong \mathcal{B}_q$ and $\mathcal{B}_p(c) \cong \mathcal{B}_p$ at each $c \in \mathcal{C}$.

A.2 Agents and Multi-Agent Systems

Agent

An agent $\mathcal{A}^i$ is a pair of local sections over a domain $\mathcal{U}_i \subset \mathcal{C}$:

$$\mathcal{A}^i = (\sigma_q^i, \sigma_p^i), \quad (3)$$

where $\sigma_q^i : \mathcal{U}_i \rightarrow \mathcal{E}_q$ and $\sigma_p^i : \mathcal{U}_i \rightarrow \mathcal{E}_p$.

We write $q_i(c) := \sigma_q^i(c) \in \mathcal{B}_q(c)$ for the belief and $s_i(c) := \sigma_p^i(c) \in \mathcal{B}_p(c)$ for the model at base point c . The symbol p_i is reserved in the main paper for the belief-channel base prior $p_i(k_i) \in \mathcal{B}_q$ (a probability density on the belief fiber, see main paper Eq. ??); the section σ_p^i on the model fiber $\mathcal{B}_p$ defined here is therefore named s_i to avoid collision.

Multi-Agent System

A multi-agent system $\mathcal{M}$ over $\mathcal{C}$ is a collection of agents indexed by $\mathcal{I}$:

$$\mathcal{M} = \{\mathcal{A}^i = (\sigma_q^i, \sigma_p^i)\}_{i \in \mathcal{I}}. \quad (4)$$

Agents generally overlap on intersections $\mathcal{U}_i \cap \mathcal{U}_j$.

Meta-Agent and Epistemic Death

A meta-agent is a multi-agent system whose component agents share identical section values on their overlap:

$$q_i(c) = q_j(c), \quad s_i(c) = s_j(c) \quad \text{for } c \in \mathcal{U}_i \cap \mathcal{U}_j. \quad (5)$$

A set of agents is epistemically dead if they identically share both beliefs and models. While constituent agents of a meta-agent may be epistemically dead, the meta-agent itself need not be; such agents can be integrated out, yielding coarse-grained higher-order entities and a route toward emergence.

A.2.1 HIERARCHICAL META-AGENT EMERGENCE AND CROSS-SCALE COUPLING

The gauge-theoretic framework naturally supports the emergence of meta-agents through coarse-graining (Kadanoff, 1966; Wilson and Kogut, 1974). A meta-agent $\mathcal{M}^{(1)}$ is a set of agents $\{i \in I_{\mathcal{M}}\}$ satisfying:

$$q_i = \Omega_{ij} q_j \quad \forall i, j \in I_{\mathcal{M}} \quad (\text{belief consensus}), \quad (6)$$

$$s_i = \tilde{\Omega}_{ij} s_j \quad \forall i, j \in I_{\mathcal{M}} \quad (\text{model consensus}). \quad (7)$$

When these conditions hold, the agents are epistemically dead in that they share identical beliefs and models after accounting for gauge transformations and no longer evolve without continual observations. The meta-agent can be described by a single representative renormalized state $(q_{\mathcal{M}}, s_{\mathcal{M}})$. The cross-scale prior-propagation construction $p_i^{(\zeta)} = \Omega_{i,I}[q_I^{(\zeta+1)}]$ developed in the companion paper (Dennis, 2025) treats priors as transported shadows of the meta-agent’s posterior at the next-higher scale. The gauge-theoretic transformer of the main paper treats priors as primitive boundary data (the per-token PriorBank, see main paper Section ??); the hierarchical formulation sketched in this sub-subsection is the framework-level scaffold that the companion paper exercises in detail, and is not invoked by the main paper’s single-scale transformer derivations.

Meta-agents at scale $\zeta = 0$ (individual agents) can form meta-agents at scale $\zeta = 1$ (groups), which can further coalesce into meta-agents at scale $\zeta = 2$ (communities), and so on, creating a hierarchical scale structure:

$$\text{Agents}^{(0)} \xrightarrow{\text{consensus}} \text{Meta-agents}^{(1)} \xrightarrow{\text{consensus}} \text{Meta}^{(2)} \xrightarrow{\text{consensus}} \dots$$

Figure 1: Hierarchical emergence through consensus at successive scales ζ .

At each scale, the effective free energy takes the same form as the full two-channel variational free energy (66), with agents at scale ζ replaced by meta-agents at scale $\zeta + 1$, coupling constants renormalized according to

$$\beta_{ij}^{(\zeta+1)} = f_{\beta}(\{\beta_{kl}^{(\zeta)}\}), \quad \gamma_{ij}^{(\zeta+1)} = f_{\gamma}(\{\gamma_{kl}^{(\zeta)}\}),$$

and effective gauge frames given by $\phi_{\mathcal{M}}^{(\zeta+1)} = \text{average}(\{\phi_i^{(\zeta)}\})$. This is the gauge-theoretic analogue of renormalization group flow in statistical field theory (Anderson and Rosenfeld, 1988; Wilson and Kogut, 1974; García-Pérez et al., 2018). Cross-scale couplings Λ_{ζ}^{ζ} , mediate interactions between agents at different hierarchical levels via the cross-scale transport operators defined below.

Whether this hierarchical structure maps onto multi-layer transformer architectures (where each layer constitutes a “scale”), or onto distinct organizational levels within a single attention layer, remains an open question for future investigation.

A.3 Bundle Morphisms and Transport Operators

Via standard horizontal lifting from the principal bundle $\mathcal{N}$ to the associated bundles, we obtain a hierarchy of morphisms and induced connections:

Intra-bundle transport operators $\Omega_{ij}^{(q)} : \Gamma(\mathcal{B}_q) \rightarrow \Gamma(\mathcal{B}_q)$ (belief-to-belief) and $\Omega_{ij}^{(p)} : \Gamma(\mathcal{B}_p) \rightarrow \Gamma(\mathcal{B}_p)$ (model-to-model) act within a single bundle. Cross-scale transport operators $\Lambda_{s'}^{s'} : \Gamma^s(\mathcal{B}_q) \rightarrow \Gamma^{s'}(\mathcal{B}_q)$ (beliefs across scales) and $\tilde{\Lambda}_{s'}^{s'} : \Gamma^s(\mathcal{B}_p) \rightarrow \Gamma^{s'}(\mathcal{B}_p)$ (models across

scales) connect different hierarchical levels. Inter-bundle morphisms $\Theta_j^i : \Gamma(\mathcal{B}_q) \rightarrow \Gamma(\mathcal{B}_p)$ (belief to model) and $\tilde{\Theta}_j^i : \Gamma(\mathcal{B}_p) \rightarrow \Gamma(\mathcal{B}_q)$ (model to belief) relate the two fiber types. Finally, global bundle morphisms $\Phi : \mathcal{E}_p \rightarrow \mathcal{E}_q$ (model bundle to belief bundle) and $\tilde{\Phi} : \mathcal{E}_q \rightarrow \mathcal{E}_p$ (belief bundle to model bundle) act on the total associated bundles.

Here $\Gamma(\mathcal{B}_u)$ denotes the space of smooth sections over the base $\mathcal{C}$.

A.4 Gauge Frames and Connections

Each agent i possesses a local gauge frame field:

$$\phi_i : \mathcal{U}_i \rightarrow \mathfrak{g} = \text{Lie}(G), \quad (8)$$

which induces a local connection one-form:

$$A_\mu^{(i)}(c) = U_i^{-1}(c) \partial_\mu U_i(c), \quad (9)$$

where $U_i(c) = \exp[\phi_i(c)] \in G$. The associated field strength (gauge curvature) is:

$$F_{\mu\nu}^{(i)}(c) = \partial_\mu A_\nu^{(i)} - \partial_\nu A_\mu^{(i)} + [A_\mu^{(i)}, A_\nu^{(i)}] \in \mathfrak{g}. \quad (10)$$

When two agents overlap at $c \in \mathcal{U}_i \cap \mathcal{U}_j$, the inter-agent gauge transformation:

$$\Omega_{ij}(c) = \exp[\phi_i(c)] \exp[-\phi_j(c)] \in G \quad (11)$$

transports agent j 's representations into agent i 's frame:

$$q_j(c) \mapsto \Omega_{ij}(c) \cdot q_j(c) := \rho(\Omega_{ij}(c)) q_j(c). \quad (12)$$

On overlaps, local connections are related by:

$$A_\mu^{(i)} = \Omega_{ij} A_\mu^{(j)} \Omega_{ij}^{-1} + \Omega_{ij} \partial_\mu \Omega_{ij}^{-1}. \quad (13)$$

Scope and forward references. The present chapter establishes the bundle scaffold and the gauge-connection apparatus used throughout this appendix and the main paper. The remaining foundational machinery on which the main paper's derivations build is developed in subsequent chapters of this appendix: the Gaussian KL closed form under $\text{GL}(K)$ gauge transport and the covariance-channel fixed-point dynamics are derived in Section B; the gauge-frame gradient and the differential of the matrix exponential are derived in Section C; the natural-gradient descent on the Gaussian manifold and the symmetric positive-definite manifold retraction are derived in Section D. The variational free energy functional itself, its E-step / M-step decomposition, and the softmax- β derivation from a mixture-of-sources model are developed in the main paper at Sections ?? and ?. The $\text{GL}(K)$ -invariance of the Gaussian KL divergence is stated and proved in the main paper at Theorem ??; this appendix uses it without re-proof.

Appendix B. Covariance Dynamics and Equilibrium Analysis

B.1 Covariance Gradient of the Generalized Free Energy

Here we derive the gradient of the single-agent free energy $\mathcal{F}_i$ with respect to the covariance Σ_i for agent i ; a well known result in information geometry. For brevity we work with the *entropy-suppressed surrogate* $\sum_j \beta_{ij} D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$ (i.e. holding the attention distribution β fixed and treating the alignment energy at this β); the canonical free energy of the main text adds the $\tau \beta_{ij} \log(\beta_{ij}/\pi_{ij})$ entropy term to make the softmax form of β a stationary point. The covariance gradient is identical under both forms because the attention entropy does not depend on Σ_i .

The free energy (surrogate form) is

$$\mathcal{F}_i = D_{\text{KL}}(q_i \| p_i) + \sum_{j \neq i} \beta_{ij} D_{\text{KL}}(q_i \| \Omega_{ij} q_j) - \mathbb{E}_{q_i}[\log p(o_i | k_i)], \quad (14)$$

where $q_i = \mathcal{N}(\mu_i, \Sigma_i)$, $p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i})$, and $\sum_j \beta_{ij} = 1$ by construction.

B.1.1 GAUSSIAN KL DIVERGENCE UNDER $\text{GL}(K)$ GAUGE TRANSPORT

The alignment term $D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$ involves KL divergences between Gaussians related by gauge transport $\Omega_{ij} = e^{\phi_i} e^{-\phi_j} \in \text{GL}^+(K)$. The standard Gaussian KL divergence (Bishop, 2006a; Murphy, 2012; Cover and Thomas, 2006) is

$$D_{\text{KL}}(\mathcal{N}(\mu_1, \Sigma_1) \| \mathcal{N}(\mu_2, \Sigma_2)) = \frac{1}{2} \left[\log \frac{|\Sigma_2|}{|\Sigma_1|} + \text{tr}(\Sigma_2^{-1} \Sigma_1) \right] \quad (15)$$

$$+ (\mu_2 - \mu_1)^\top \Sigma_2^{-1} (\mu_2 - \mu_1) - d \Big]. \quad (16)$$

Differentiating w.r.t. Σ_1 (holding μ_1, μ_2, Σ_2 fixed), using the standard identities $\partial \log |\Sigma| / \partial \Sigma = \Sigma^{-\top}$ and $\partial \text{tr}(A \Sigma) / \partial \Sigma = A^\top$ (Petersen and Pedersen, 2012), gives

$$\frac{\partial D_{\text{KL}}}{\partial \Sigma_1} = \frac{1}{2} [-\Sigma_1^{-1} + \Sigma_2^{-1}]. \quad (17)$$

The push-forward $\Omega_{ij} q_j = \mathcal{N}(\Omega_{ij} \mu_j, \Omega_{ij} \Sigma_j \Omega_{ij}^\top)$ used below is the standard linear-transformation rule for a Gaussian random variable under $\Omega \in \text{GL}(K)$ (Bishop, 2006a, §2.3.3). Applying the above derivative to each term in $\mathcal{F}_i$:

$$\frac{\partial}{\partial \Sigma_i} D_{\text{KL}}(q_i \| p_i) = \frac{1}{2} [-\Sigma_i^{-1} + \Sigma_{p,i}^{-1}], \quad (18)$$

$$\frac{\partial}{\partial \Sigma_i} D_{\text{KL}}(q_i \| \Omega_{ij} q_j) = \frac{1}{2} [-\Sigma_i^{-1} + (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1}]. \quad (19)$$

The observation term contributes a model-dependent correction to the covariance gradient (e.g., $\frac{1}{2} R^{-1}$ for Gaussian observations with noise covariance R). In the high-precision regime where $\Sigma_i^{-1} \gg R^{-1}$, this $O(1)$ correction is negligible compared to the $O(\Sigma_i^{-1})$

alignment and prior terms. The alignment term $\sum_j \beta_{ij} D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$ requires the product rule when differentiating the attention-weighted energy (the autograd objective), since $\beta_{ij} = \text{softmax}_j(-D_{\text{KL}}/\tau)$ depends on Σ_i through the KL divergence. Thus

$$\boxed{\begin{aligned} \frac{\partial \mathcal{F}_i}{\partial \Sigma_i} = & \frac{1}{2} \left[-2\Sigma_i^{-1} + \Sigma_{p,i}^{-1} + \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \right] \\ & + \sum_j \frac{\partial \beta_{ij}}{\partial \Sigma_i} D_{\text{KL}}(q_i \| \Omega_{ij} q_j). \end{aligned}} \quad (20)$$

The first line contains the “direct” gradient terms (differentiating each K_{ij} at fixed β_{ij}). The second line is the attention weight gradient, analogous to the $\partial \beta_{ij} / \partial \mu_i$ term in the mean gradient. This $\partial \beta / \partial \Sigma$ term arises from differentiating the attention-weighted energy $\sum_j \beta_{ij} E_{ij}$ via the product rule; it is absent from the gradient of the reduced free energy $\mathcal{F}_{\text{red}} = -\tau \log Z_i + \dots$ by the envelope theorem (see main text, Section 3.5). From the softmax structure, $\sum_j \partial \beta_{ij} / \partial \Sigma_i = 0$, so this correction vanishes when the KL divergences are uniform across neighbors, and also in both the $\tau \rightarrow 0$ and $\tau \rightarrow \infty$ limits.

The coefficient of Σ_i^{-1} is $-(1 + \sum_j \beta_{ij}) = -2$, since the entropy of q_i contributes $-\frac{1}{2} \Sigma_i^{-1}$ to both the prior KL (once, with the self-coupling weight α_i on the prior term held at its canonical default $\alpha_i = 1$; see main paper Section ?? for the state-dependent generalization) and the attention-weighted alignment KL (once, weighted by $\sum_j \beta_{ij} = 1$).

B.2 Fixed-Point Equation and Symmetric Solution

At equilibrium, we set $\partial \mathcal{F}_i / \partial \Sigma_i = 0$. Since $\sum_j \partial \beta_{ij} / \partial \Sigma_i = 0$, the attention weight correction term vanishes whenever the KL divergences $D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$ are approximately uniform across neighbors, and also in both the hard-attention ($\tau \rightarrow 0$) and uniform-attention ($\tau \rightarrow \infty$) limits. In these regimes, the equilibrium condition reduces to

$$\Sigma_i^{-1} = \frac{1}{2} \left[\Sigma_{p,i}^{-1} + \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \right]. \quad (21)$$

This is a matrix-valued fixed-point equation coupling all agents: each agent’s precision is the β -weighted combination of its own prior precision and the transported neighbor precisions. At intermediate temperatures, the attention weight correction provides an additional contribution that couples precision dynamics to the variance of KL divergences across neighbors.

B.2.1 HOMOGENEOUS LIMIT

In the homogenous limit we assume

- (i) all agents are identical, so $\Sigma_i = \Sigma_\infty$ for all i ,
- (ii) $\Omega_{ij} \approx I$ (weak misalignment),
- (iii) $\Sigma_{p,i} = \Sigma_0$ (shared prior).

Then (21) becomes

$$\Sigma_\infty^{-1} = \frac{1}{2} [\Sigma_0^{-1} + \Sigma_\infty^{-1}] \implies \Sigma_\infty = \Sigma_0. \quad (22)$$

Hence, in a perfectly symmetric population, the equilibrium covariance reproduces the shared prior.

B.2.2 ALIGNMENT-DOMINATED REGIME

Although $\sum_j \beta_{ij} = 1$, the effective strength of alignment is controlled by the parameter τ in

$$\beta_{ij} = \frac{\exp \left[-\frac{1}{\tau} D_{\text{KL}}(q_i \| \Omega_{ij} q_j) \right]}{\sum_k \exp \left[-\frac{1}{\tau} D_{\text{KL}}(q_i \| \Omega_{ik} q_k) \right]}. \quad (23)$$

As $\tau \rightarrow 0$, β_{ij} becomes sharply peaked on whichever neighbor j best agrees (after transport). Small τ alone sharpens the β distribution but does not change the coefficient on Σ_i^{-1} in the fixed-point equation (21), which remains $\frac{1}{2}$. To obtain a coefficient of 1 on the alignment term, one requires the state-dependent prior coupling $\alpha_i \ll 1$ (cf. main text Section ??), which changes the coefficient from $-(1 + \alpha_i)$ to approximately -1 . In the alignment-dominated regime where both $\alpha_i \ll 1$ and τ is small, the fixed point becomes:

$$\boxed{\Sigma_i^{-1} \approx \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} = \langle (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \rangle_\beta.} \quad (24)$$

Here $\langle \cdot \rangle_\beta$ denotes a β -weighted expectation over neighbors.

In this regime, agent i 's precision matrix becomes the β -weighted average of its neighbors' transported precisions. The two conditions play complementary roles: small α_i suppresses the prior term and changes the coefficient to unity, while small τ concentrates β on the most aligned neighbors. This regime is studied as a theoretical limit; the canonical experimental configuration of the main paper uses $\alpha_i = 1$, in which case the prior term remains present at coefficient $\frac{1}{2}$ and the fixed-point form (21) applies.

When, additionally, all agents already have approximately equal covariances and the transports are near-identity, i.e. $\Sigma_i \approx \Sigma_j$ and $\Omega_{ij} \approx I$, then (24) implies

$$\Sigma_i^{-1} \approx \Sigma_j^{-1} \implies \Sigma_i \approx \Omega_{ij} \Sigma_j \Omega_{ij}^\top, \quad (25)$$

justifying the alignment assumption used in the main text.

B.2.3 GRADIENT FLOW DYNAMICS

Finally, consider the gradient flow

$$\frac{d\Sigma_i}{dt} = -\eta_\Sigma \frac{\partial \mathcal{F}_i}{\partial \Sigma_i}, \quad \eta_\Sigma > 0. \quad (26)$$

Local stability of the equilibrium (21) follows from the positive-definiteness of the Hessian. For the Gaussian KL terms $D_{\text{KL}}(\mathcal{N}(\mu_1, \Sigma_1) \| \mathcal{N}(\mu_2, \Sigma_2))$ viewed as a function of Σ_1 at fixed Σ_2 , the second-variation acts on symmetric tangent vectors $H \in \text{Sym}(K)$ as

$$\frac{\partial^2 D_{\text{KL}}}{\partial \Sigma_1 \partial \Sigma_1} \left[H, H \right] = \frac{1}{2} \text{tr}(\Sigma_1^{-1} H \Sigma_1^{-1} H) = \frac{1}{2} \|\Sigma_1^{-1/2} H \Sigma_1^{-1/2}\|_F^2 \geq 0, \quad (27)$$

with equality iff $H = 0$. The Hessian depends only on Σ_1 (the $\text{tr}(\Sigma_2^{-1} \Sigma_1)$ term is linear in Σ_1 and contributes no Σ_2 -dependent second derivative), and is manifestly positive definite for $\Sigma_1 \succ 0$. The full Hessian of $\mathcal{F}_i$ extends to a β -weighted convex combination of single-KL Hessians at the joint (β, Σ) stationary point; the envelope-theorem identity of Eq. (??) (main paper) suppresses the $\partial\beta/\partial\Sigma$ contributions at the fixed point.

Hence the covariance alignment fixed-point is an attractor of the variational dynamics under the standing assumptions $\sum_j \beta_{ij} = 1$, Ω_{ij} invertible, and the observation precision R^{-1} negligible relative to the inter-agent coupling (cf. Eq. (20)).

Under these assumptions the alignment $\Sigma_i \approx \Omega_{ij} \Sigma_j \Omega_{ij}^\top$ is consistent with the variational dynamics; we do not claim a contractive proof of global uniqueness here.

Appendix C. Gauge Frame Gradients

We derive the gradient of the variational free energy $\mathcal{F}$ with respect to the gauge frame parameters $\phi_i \in \mathfrak{g}$. Unlike the covariance gradients (Appendix B), gauge frame gradients require the differential of the matrix exponential map, and the resulting expressions differ for compact (e.g., $\text{SO}(N)$) and general ($\text{GL}(K)$) gauge groups.

C.1 Structure of the Gauge Frame Gradient

Convention. Throughout this appendix and Appendix D, the components $\partial\mathcal{F}/\partial\phi_i^a$ of the gauge-frame gradient in the generator basis $\{T_a\}$ are defined via the *right-trivialised* differential of the matrix exponential, $D_{\phi_i}(\exp)[T_a] = Q_a^{R,(i)} \exp(\phi_i)$ with $Q_a^{R,(i)} \equiv \text{dexp}_{\phi_i}(T_a)$, so that $\partial\Omega_{ij}/\partial\phi_i^a = Q_a^{R,(i)} \Omega_{ij}$. The implementation computes these components by automatic differentiation through PyTorch’s `torch.matrix_exp`, which returns the Fréchet derivative $D_\phi(\exp)[T_a]$; the right-trivialised generator is then recovered as $Q_a^R = D_\phi(\exp)[T_a] \cdot e^{-\phi}$. The same components are fed to the Cartan-modified preconditioner $\tilde{g}_{ab}$ of Section C.5 (Eq. 47). The left-trivialised form $\partial/\partial\phi^a \exp(X) = \exp(X) \cdot Q_a^L$ with $Q_a^L = \text{Ad}_{\exp(-\phi)}(Q_a^R)$ is mathematically equivalent and is sometimes written in Appendix D for didactic comparison; the reference implementation uses the right-trivialised convention throughout.

The gauge frames enter the free energy exclusively through the parallel transport operators $\Omega_{ij} = \exp(\phi_i) \exp(-\phi_j)$. Since neither the belief-prior term $D_{\text{KL}}(q_i \| p_i)$ nor the observation likelihood $-\mathbb{E}_{q_i}[\log p(o_i | k_i)]$ depends on ϕ_i , the complete autograd gradient (differentiating the attention-weighted energy $\sum_j \beta_{ij} K_{ij}$ via the product rule; see main text Section 3.5 for the distinction between autograd and reduced-free-energy gradients) is:

$$\frac{\partial \mathcal{F}}{\partial \phi_i} = \sum_j \left[\frac{\partial \beta_{ij}}{\partial \phi_i} K_{ij} + \beta_{ij} \frac{\partial K_{ij}}{\partial \phi_i} \right] + \sum_k \left[\beta_{ki} \frac{\partial K_{ki}}{\partial \phi_i} + \sum_l \frac{\partial \beta_{kl}}{\partial \phi_i} K_{kl} \right], \quad (28)$$

where $K_{ij} \equiv D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$. The first sum captures agent i aligning to others, with the product rule applied to $\beta_{ij} K_{ij}$. The second sum captures others aligning to i : changing ϕ_i shifts K_{ki} through $\Omega_{ki} = \exp(\phi_k) \exp(-\phi_i)$, which both directly adjusts the align-

ment cost (the $\beta_{ki}\partial K_{ki}/\partial\phi_i$ term) and redistributes all of agent k 's attention weights (the $\sum_l(\partial\beta_{kl}/\partial\phi_i)K_{kl}$ term). Since only K_{ki} depends on ϕ_i in agent k 's softmax, the latter simplifies to:

$$\sum_l \frac{\partial\beta_{kl}}{\partial\phi_i} K_{kl} = -\frac{\beta_{ki}}{\tau} \frac{\partial K_{ki}}{\partial\phi_i} \left(K_{ki} - \sum_l \beta_{kl} K_{kl} \right). \quad (29)$$

This correction vanishes when K_{ki} equals agent k 's average divergence $\bar{K}_k = \sum_l \beta_{kl} K_{kl}$, or when $\beta_{ki} \rightarrow 0$ (agent k does not attend to i).

The coupling weight gradient in the first sum follows from the softmax structure:

$$\frac{\partial\beta_{ij}}{\partial\phi_i} = -\frac{\beta_{ij}}{\tau} \left[\frac{\partial K_{ij}}{\partial\phi_i} - \sum_k \beta_{ik} \frac{\partial K_{ik}}{\partial\phi_i} \right]. \quad (30)$$

C.2 Differential of the Matrix Exponential

All KL gradient terms require the derivative of the transport operator with respect to the gauge frame. For $\Omega_{ij} = \exp(\phi_i) \exp(-\phi_j)$:

$$\frac{\partial\Omega_{ij}}{\partial\phi_i} = \text{dexp}_{\phi_i}(\cdot) \cdot \exp(-\phi_j), \quad (31)$$

$$\frac{\partial\Omega_{ij}}{\partial\phi_j} = -\exp(\phi_i) \cdot \text{dexp}_{-\phi_j}(\cdot) \cdot \exp(-\phi_j), \quad (32)$$

where $\text{dexp}_\phi : \mathfrak{g} \rightarrow \mathfrak{g}$ denotes the differential of the exponential map at ϕ . For a direction $\xi \in \mathfrak{g}$, the differential is given by (Gallier and Quaintance, 2020; Hall, 2015):

$$\text{dexp}_\phi(\xi) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} \text{ad}_\phi^n(\xi) = \frac{e^{\text{ad}_\phi} - I}{\text{ad}_\phi}(\xi), \quad (33)$$

where $\text{ad}_\phi(\xi) = [\phi, \xi] = \phi\xi - \xi\phi$ is the adjoint representation. Equation (33) defines the right-trivialized differential, related to the Fréchet derivative (directional derivative in matrix space) by $D_\phi(\exp)[\xi] = \text{dexp}_\phi(\xi) \cdot e^\phi$. The $\text{GL}(K)$ specialization below uses the Fréchet derivative directly via its integral representation (Higham, 2008, Ch. 10).

C.2.1 $\text{SO}(N)$ SPECIALIZATION.

For $\text{SO}(N)$, the gauge frame is expanded in skew-symmetric generators: $\phi = \sum_a \phi^a T_a$ where $T_a \in \mathfrak{so}(N)$, and the gradient is computed component-wise as $\partial\mathcal{F}/\partial\phi^a$. For $\text{SO}(3)$ with $\|\phi\| = \theta$, the differential admits a closed form via the Rodrigues formula:

$$\text{dexp}_\phi(T_a) = T_a + c_1(\theta)[\phi, T_a] + c_2(\theta)[\phi, [\phi, T_a]], \quad (34)$$

with coefficients:

$$c_1(\theta) = \frac{1 - \cos \theta}{\theta^2}, \quad c_2(\theta) = \frac{\theta - \sin \theta}{\theta^3}. \quad (35)$$

For $\theta < \epsilon$ (numerically $\epsilon \sim 10^{-4}$), Taylor expansions $c_1 \approx \frac{1}{2} - \frac{\theta^2}{24}$ and $c_2 \approx \frac{1}{6} - \frac{\theta^2}{120}$ provide stable evaluation.

C.2.2 GL(K) SPECIALIZATION.

For GL(K), the gauge frame $\phi \in \mathfrak{gl}(K) \cong \mathbb{R}^{K \times K}$ is a general $K \times K$ matrix. The right-trivialised differential of the exponential does not simplify to a Rodrigues-like formula. The full Fréchet derivative $D_\phi(\exp)[\xi] = \text{dexp}_\phi(\xi) \cdot e^\phi$ admits an integral representation (Higham, 2008, Eq. 10.15):

$$D_\phi(\exp)[\xi] = \int_0^1 e^{t\phi} \xi e^{(1-t)\phi} dt, \quad (36)$$

or equivalently through the $(2K \times 2K)$ block matrix identity (Higham, 2008, Algorithm 10.27):

$$\exp \begin{pmatrix} \phi & \xi \\ 0 & \phi \end{pmatrix} = \begin{pmatrix} e^\phi & D_\phi(\exp)[\xi] \\ 0 & e^\phi \end{pmatrix}. \quad (37)$$

These give the full Fréchet derivative directly; the right-trivialised generator $\text{dexp}_\phi(\xi)$ of Eq. (33) is recovered as $\text{dexp}_\phi(\xi) = D_\phi(\exp)[\xi] \cdot e^{-\phi}$. In practice, automatic differentiation through `torch.matrix_exp` provides the GL(K) differential without explicit computation of (36) or (37).

C.3 KL Gradient Through Transport

The KL divergence $K_{ij} = D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$ between $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and the transported belief $\Omega_{ij} q_j = \mathcal{N}(\Omega_{ij} \mu_j, \Omega_{ij} \Sigma_j \Omega_{ij}^\top)$ depends on ϕ_i through Ω_{ij} . Using the Gaussian KL (16) and the chain rule:

$$\frac{\partial K_{ij}}{\partial \phi_i^a} = \frac{\partial}{\partial \phi_i^a} \left\{ \frac{1}{2} \left[\log \frac{|\Omega_{ij} \Sigma_j \Omega_{ij}^\top|}{|\Sigma_i|} + \text{tr} \left((\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \Sigma_i \right) + \Delta \mu_{ij}^\top (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \Delta \mu_{ij} - K \right] \right\}, \quad (38)$$

where $\Delta \mu_{ij} = \mu_i - \Omega_{ij} \mu_j$. Let $\tilde{\Sigma}_{ij} \equiv \Omega_{ij} \Sigma_j \Omega_{ij}^\top$ and $\tilde{\mu}_{ij} \equiv \Omega_{ij} \mu_j$ denote the transported statistics. The derivative has three contributions:

Mean term. From the Mahalanobis distance:

$$\frac{\partial}{\partial \phi_i^a} \left[(\mu_i - \tilde{\mu}_{ij})^\top \tilde{\Sigma}_{ij}^{-1} (\mu_i - \tilde{\mu}_{ij}) \right] = -2(\mu_i - \tilde{\mu}_{ij})^\top \tilde{\Sigma}_{ij}^{-1} \frac{\partial \tilde{\mu}_{ij}}{\partial \phi_i^a} + (\text{terms from } \partial \tilde{\Sigma}_{ij}^{-1} / \partial \phi_i^a). \quad (39)$$

Trace term. From the precision-covariance product:

$$\frac{\partial}{\partial \phi_i^a} \text{tr}(\tilde{\Sigma}_{ij}^{-1} \Sigma_i) = -\text{tr} \left(\tilde{\Sigma}_{ij}^{-1} \frac{\partial \tilde{\Sigma}_{ij}}{\partial \phi_i^a} \tilde{\Sigma}_{ij}^{-1} \Sigma_i \right). \quad (40)$$

Log-determinant term. From the log-determinant ratio:

$$\frac{\partial}{\partial \phi_i^a} \log |\tilde{\Sigma}_{ij}| = \text{tr} \left(\tilde{\Sigma}_{ij}^{-1} \frac{\partial \tilde{\Sigma}_{ij}}{\partial \phi_i^a} \right). \quad (41)$$

Transport derivatives. The derivatives of the transported statistics with respect to ϕ_i^a are:

$$\frac{\partial \tilde{\mu}_{ij}}{\partial \phi_i^a} = \frac{\partial (\Omega_{ij} \mu_j)}{\partial \phi_i^a} = Q_a^{(i)} \Omega_{ij} \mu_j, \quad (42)$$

$$\frac{\partial \tilde{\Sigma}_{ij}}{\partial \phi_i^a} = Q_a^{(i)} \Omega_{ij} \Sigma_j \Omega_{ij}^\top + \Omega_{ij} \Sigma_j \Omega_{ij}^\top (Q_a^{(i)})^\top, \quad (43)$$

where $Q_a^{(i)} \equiv \text{dexp}_{\phi_i}(T_a)$ is the right-trivialised differential of the exponential map at ϕ_i in the direction of generator T_a , defined by the identity $D_{\phi_i}(\exp)[T_a] = Q_a^{(i)} \exp(\phi_i)$. The chain rule applied to $\Omega_{ij} = \exp(\phi_i) \exp(-\phi_j)$ then gives $\partial \Omega_{ij} / \partial \phi_i^a = D_{\phi_i}(\exp)[T_a] \exp(-\phi_j) = Q_a^{(i)} \Omega_{ij}$, which is the form used above. (An equivalent expression in terms of the Fréchet derivative is $\partial \Omega_{ij} / \partial \phi_i^a = D_{\phi_i}(\exp)[T_a] \exp(-\phi_j)$; both conventions appear in the implementation, but the numerical retraction uses the right-trivialised form for consistency with the Lie-algebra update (44).)

C.4 Retraction and Numerical Considerations

Lie algebra updates. Since $\phi_i \in \mathfrak{g}$ (a vector space), gradient descent proceeds in the Lie algebra directly:

$$\phi_i \leftarrow \phi_i - \eta_\phi \frac{\partial \mathcal{F}}{\partial \phi_i}, \quad (44)$$

where η_ϕ is the gauge frame learning rate. No metric correction (Fisher-Rao or otherwise) is applied: the exponential map $\exp : \mathfrak{g} \rightarrow G$ provides natural coordinates for the Lie group, and the gradient naturally lives in $\mathfrak{g}$.

SO(N) retraction. For compact groups such as $\text{SO}(N)$, the exponential map is periodic. To ensure unique parameterization and avoid branch cuts, we retract to the principal ball $\|\phi\| < \pi - \epsilon$ (with $\epsilon \sim 10^{-2}$) after each update. If $\|\phi\| \geq \pi$, the angle is wrapped modulo 2π and, if necessary, the axis is flipped via antipodal identification ($\phi \rightarrow -\phi$ when $\theta > \pi$), exploiting the fact that $\exp(\phi) = \exp(\phi - 2\pi \hat{n})$ for $\text{SO}(3)$.

GL(K) updates. For $\text{GL}(K)$, a single matrix exponential $\exp : \mathfrak{gl}(K) \rightarrow \text{GL}^+(K)$ is not surjective onto the identity component (Culver, 1966), but the pairwise transport $\Omega_{ij} = \exp(\phi_i) \exp(-\phi_j)$ covers all of $\text{GL}^+(K)$ via polar decomposition (Higham, 2008, §8): for any $A \in \text{GL}^+(K)$ with polar decomposition $A = UP$ ($U \in \text{SO}(K)$, $P \in \text{Sym}_{++}(K)$), the exponential map restricts to a surjection $\mathfrak{so}(K) \twoheadrightarrow \text{SO}(K)$ on the compact factor and a bijection $\text{Sym}(K) \xrightarrow{\sim} \text{Sym}_{++}(K)$ on the symmetric factor, so choosing ϕ_i with $\exp(\phi_i) = UP^{1/2}$ and ϕ_j with $\exp(-\phi_j) = P^{1/2}$ realizes $\Omega_{ij} = A$. No retraction is required beyond standard gradient clipping for numerical stability. In our implementation, automatic differentiation through PyTorch’s `matrix_exp` handles the $\text{GL}(K)$ differential implicitly.

C.5 Gauge Frame Preconditioning for $\text{GL}(K)$

While the basic update (44) uses the Euclidean gradient in the Lie algebra, the non-compact structure of $\text{GL}(K)$ introduces a numerical asymmetry that motivates gauge-aware preconditioning. The Lie algebra decomposes via the Cartan involution $\theta(X) = -X^\top$ as:

$$\mathfrak{gl}(K) = \underbrace{\mathfrak{so}(K)}_{\text{compact (skew-symmetric)}} \oplus \underbrace{\text{Sym}(K)}_{\text{non-compact (symmetric)}} \oplus \underbrace{\mathbb{R}}_{\text{center (trace)}}, \quad (45)$$

where $\mathfrak{so}(K) = \{X : X + X^\top = 0\}$ generates rotations and $\text{Sym}(K) = \{X : X = X^\top, \text{tr } X = 0\}$ generates volume-preserving shears. These two subspaces have qualitatively different gradient geometry: for the compact directions, $\|d\exp_\phi(T_a)\| = O(1)$ uniformly in $\|\phi\|$, while for the symmetric directions, $\|d\exp_\phi(T_a)\| \sim \exp(\|\phi\|)$. Euclidean gradient descent therefore systematically over-steps in the non-compact directions relative to the compact ones, leading to numerical instability for $\text{GL}(K)$ gauge frames.

We implement four preconditioning strategies of increasing geometric fidelity:

- 1. Norm clipping (baseline).** Simple gradient norm clipping $\tilde{g} = g \cdot \min(1, c/\|g\|)$ with threshold c . No geometric awareness; treats all Lie algebra directions uniformly.
- 2. Cartan preconditioning.** Decompose the gradient into compact and non-compact components and selectively dampen the latter. Given Lie algebra generators $\{T_a\}_{a=1}^{n_{\text{gen}}}$ with Gram matrix $G_{ab} = \text{tr}(T_a^\top T_b)$ and symmetric inner product $S_{ab} = \text{tr}(T_a T_b)$, the projector onto the symmetric subspace is:

$$P_{\text{sym}} = \frac{1}{2}G^{-1}(G + S), \quad (46)$$

and the preconditioner is $M = I - (1 - \lambda_{\text{sym}})P_{\text{sym}}$ with dampening factor $\lambda_{\text{sym}} \in (0, 1)$ (typically 0.1, corresponding to $10\times$ dampening of symmetric directions). The preconditioned gradient is $\tilde{\nabla}_\phi \mathcal{F} = M \cdot \nabla_\phi \mathcal{F}$.

- 3. Cartan-involution-modified bilinear form (compact-form metric).** The Killing form of $\mathfrak{gl}(K)$ is $B(X, Y) = 2K \text{tr}(XY) - 2\text{tr}(X)\text{tr}(Y)$. On the trace-free subalgebra $\mathfrak{sl}(K) = \ker(\text{tr})$ it reduces to $B|_{\mathfrak{sl}(K)}(X, Y) = 2K \text{tr}(XY)$, and on the center $\mathbb{R} \cdot I$ it vanishes identically. The Killing form is *indefinite* on real $\mathfrak{sl}(K)$ for $K \geq 2$: under the Cartan decomposition $\mathfrak{sl}(K) = \mathfrak{so}(K) \oplus \text{Sym}_0(K)$ (skew $\oplus$ traceless-symmetric), B is *negative* on $\mathfrak{so}(K)$ and *positive* on $\text{Sym}_0(K)$ (Hall, 2015; Gallier and Quaintance, 2020).

The metric used in this paper is therefore not the Killing form itself but the Cartan-involution-modified bilinear form $g(X, Y) = -B(X, \theta(Y))$ with $\theta(X) = -X^\top$ (Knapp, 2002, Prop. 1.93). Equivalently, in the generator basis:

$$\tilde{g}_{ab} = 2K \text{tr}(T_a^\top T_b) - 2\text{tr}(T_a)\text{tr}(T_b), \quad (47)$$

which is positive definite on $\mathfrak{sl}(K)$ (the transpose flips the sign on $\mathfrak{so}(K)$, since $\text{tr}(T_a^\top T_a) = -\text{tr}(T_a^2) = \|T_a\|_F^2 > 0$ for skew T_a) and degenerate on the center $\mathbb{R} \cdot I$ (regularized by $\tilde{g} \rightarrow \tilde{g} + \epsilon I$ in practice). This is the compact-form metric familiar from Riemannian geometry on compact Lie groups; on the compact subgroup $\text{SO}(K)$ it agrees with $-B$ up to scale. The natural gradient $\tilde{g}^{-1}\nabla_\phi \mathcal{F}$ has no free parameters: the metric is determined

entirely by the algebra structure. This treatment of the compact/non-compact asymmetry assigns positive “lengths” to both rotations and shears, in contrast to the genuine Killing form which assigns opposite signs.

4. Pullback natural gradient (position-dependent). The most geometrically principled approach uses the pullback of the Frobenius inner product on $\text{Mat}(K)$ through the exponential map. The Fréchet derivative of $\exp$ at $\phi \in \mathfrak{g}$ in direction T_a admits the right-trivialised form (Hall, 2015, §2.7 / Theorem 5.4):

$$D_\phi(\exp)[T_a] = \Psi(\text{ad}_\phi)(T_a) \cdot \exp(\phi), \quad \Psi(z) = \frac{e^z - 1}{z} = \sum_{k=0}^{\infty} \frac{z^k}{(k+1)!}, \quad (48)$$

where $\text{ad}_\phi(X) = [\phi, X]$ is the adjoint action and $\Psi(z) = (e^z - 1)/z$ is the generating series for the right-trivialised differential; its multiplicative inverse $z/(e^z - 1)$ is the Bernoulli generating function whose Taylor coefficients are the Bernoulli numbers and which appears in $d\exp^{-1}$ and the Magnus expansion (Hall, 2015). The equivalent left-trivialised form is $D_\phi(\exp)[T_a] = \exp(\phi) \cdot ((I - e^{-\text{ad}_\phi})/\text{ad}_\phi)(T_a)$, related by the operator identity $\Psi_R(z) = e^z \Psi_L(z)$. The Frobenius pullback metric on $\mathfrak{g}$ is then:

$$\mathcal{G}_{ab}(\phi) = \langle D_\phi(\exp)[T_a], D_\phi(\exp)[T_b] \rangle_F = \text{tr}(\Psi(\text{ad}_\phi)(T_a)^\top \cdot \exp(\phi) \exp(\phi)^\top \cdot \Psi(\text{ad}_\phi)(T_b)), \quad (49)$$

where $\langle X, Y \rangle_F = \text{tr}(X^\top Y)$ is the Frobenius inner product on $\text{Mat}(K)$. At $\phi = 0$, $\exp(\phi) \exp(\phi)^\top = I$ and $\mathcal{G}_{ab}(0) = G_{ab}$ (the Gram matrix). On the compact subalgebra $\mathfrak{so}(K) \subset \mathfrak{gl}(K)$, $\exp(\phi) \in \text{SO}(K)$ so $\exp(\phi) \exp(\phi)^\top = I$ and the metric reduces to $\langle \Psi(\text{ad}_\phi)(T_a), \Psi(\text{ad}_\phi)(T_b) \rangle_F$. For non-compact (symmetric) ϕ , $\exp(\phi) \exp(\phi)^\top = \exp(2\phi)$ grows exponentially, automatically compensating the exponential amplification of $D\exp$ in the non-compact sector. The natural gradient $\mathcal{G}(\phi)^{-1} \nabla_\phi \mathcal{F}$ is position-dependent and exact (up to series truncation of Ψ), requiring the structure constants f_{ab}^c defined by $[T_a, T_b] = \sum_c f_{ab}^c T_c$.

The computational cost of each mode scales as: clipping $O(n_{\text{gen}})$, Cartan $O(n_{\text{gen}}^2)$, Killing $O(n_{\text{gen}}^2)$, pullback $O(n_{\text{gen}}^3)$ (dominated by the linear solve $\mathcal{G}^{-1}g$). For the GL(10) model ($n_{\text{gen}} = 100$), Killing form preconditioning adds negligible overhead. The pullback metric becomes advantageous when gauge frames develop large norms during training, as the position-dependent compensation prevents the gradient from overshooting in the symmetric directions where $\exp(\phi)$ amplifies exponentially.

Appendix D. Variational Gradient Descent: Implementation and Numerical Methods

D.1 Natural Gradient Descent on the Gaussian Manifold

Our implementation uses natural gradient descent for the statistical parameters, which exploits the information-geometric structure of the Gaussian manifold. Rather than treating the covariance matrices Σ_i as Euclidean variables, we recognize them as points on the manifold of symmetric positive-definite matrices, equipped with the Fisher-Rao metric (Čencov, 1982; Amari, 2016). The natural gradient at a point Σ on this manifold is obtained by

applying the inverse Fisher metric to the Euclidean gradient, yielding an intrinsic tangent vector that respects the manifold’s curved geometry. Given Euclidean gradients ∇_μ and ∇_Σ , the natural gradient projections are

$$\delta\mu = -\Sigma\nabla_\mu, \quad \delta\Sigma = -2\Sigma\text{sym}(\nabla_\Sigma)\Sigma, \quad (50)$$

where $\text{sym}(M) = \frac{1}{2}(M + M^\top)$.

The covariance update $\delta\Sigma$ is then passed to a manifold retraction (Section D.2) that whitens it into the eigenbasis of Σ :

$$B = \Lambda^{-1/2}(V^\top\delta\Sigma V)\Lambda^{-1/2}, \quad (51)$$

where $\Sigma = V\Lambda V^\top$ is the eigendecomposition. This whitening ensures that the resulting step is affine-invariant such that the same step size produces comparable effects regardless of the current scale of the covariance matrix. Following the computation of the whitened tangent B , we apply a trust-region constraint by clipping its Frobenius norm to a maximum relative step size ρ . This prevents excessively large updates that could destabilize the manifold structure or lead to ill-conditioned covariances.

D.2 Manifold Retraction for Covariance Matrices

The manifold retraction is the process of mapping the tangent vector back to a valid point on the manifold. This is performed using the affine-invariant exponential map on the manifold of symmetric positive-definite matrices (Pennec, 2006; Bhatia, 2007; Absil et al., 2008). Working in the eigenbasis of Σ_k with eigenvalues λ_j , the whitened tangent B is diagonalized $B = U\Lambda_B U^\top$ and the retraction is

$$\Sigma_{k+1} = V\Lambda^{1/2}U\exp(\tau\Lambda_B)U^\top\Lambda^{1/2}V^\top, \quad (52)$$

where τ is the step size and the component-wise exponentials $\exp(\tau\lambda_B^{(j)})$ are clipped to $|\tau\lambda_B^{(j)}| \leq 50$ to prevent floating-point overflow. This procedure is provably SPD-preserving provided Σ_k is SPD and B is symmetric, which our implementation guarantees through explicit symmetrization at multiple stages.

Additional safeguards include post-retraction sanitization via `sanitize_sigma`, which symmetrizes the result, raises on any floating-point anomalies (NaNs), and applies a spectral floor to the eigenvalues to ensure they remain above a minimum threshold $\epsilon_{\text{SPD}} = 10^{-4}$, together with a condition-number cap $\kappa_{\text{max}} = 10^4$. This spectral regularization is essential for preventing numerical collapse of the covariance matrices during prolonged gradient descent, particularly in regions where the free-energy landscape becomes very flat.

D.3 Gauge Frame Dynamics on $\text{GL}(K)$

For the gauge frame fields ϕ_i , which live in the Lie algebra $\mathfrak{gl}(K)$ (or its subalgebra $\mathfrak{so}(3)$ in the fundamental representation), a different geometric structure must be respected. These fields parameterize the local gauge frames as $U_i = \exp(\sum_a \phi_i^a T_a)$, where $\{T_a\}$ are the Lie algebra generators, and the transport operators take the form $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$. Gradients of the alignment energy $\beta_{ij}D_{\text{KL}}(q_i\|\Omega_{ij}[q_j])$ with respect to ϕ_i are obtained via the

chain rule through the transport operator: the KL gradient with respect to Ω is contracted against the differential $\partial\Omega_{ij}/\partial\phi_i^a$.

Per the convention paragraph at the head of Appendix C, all gradient components $\partial\mathcal{F}/\partial\phi_i^a$ are defined via the *right-trivialised* differential and computed by automatic differentiation through `torch.matrix_exp`. The equivalent *left-trivialised* expression is

$$\frac{\partial}{\partial\phi^a} \exp\left(\sum_b \phi^b T_b\right) = \exp(X) \cdot Q_a^L, \quad (53)$$

with $X = \sum_b \phi^b T_b$ and $Q_a^L = \text{Ad}_{\exp(-\phi)}(Q_a^R)$ (Hall, 2015); the two forms are related by $Q_a^R \exp(\phi) = \exp(\phi) \cdot Q_a^L$ and yield identical Fréchet derivatives $D_\phi(\exp)[T_a]$ when paired with the appropriate group-element factor. We display the left-trivialised form here for cross-reference and to make explicit that switching conventions away from $\phi = 0$ introduces an Ad-twist on the gradient components that the preconditioner $\tilde{g}_{ab}$ (Section C.5) would then act on; the implementation uses the right-trivialised form to compute and precondition the gradient consistently. For the $K = 3$ fundamental representation of $\mathfrak{so}(3)$, the closed-form right-trivialised expression $Q_a^R = \text{dexp}_\phi(T_a)$ of Eq. (34) (Rodrigues coefficients) is used directly. For general K , the right-trivialised generator is recovered from PyTorch’s Fréchet derivative as $Q_a^R = D_\phi(\exp)[T_a] \cdot e^{-\phi}$; no separate symmetric quadrature routine is invoked in production.

The resulting Euclidean gradient $\partial\mathcal{F}/\partial\phi^a$ can optionally be preconditioned using one of several Lie-algebraic metrics. In the transformer implementation, these include: (i) the *Cartan-involution-modified* natural gradient, which uses the bilinear form $\tilde{g}_{ab} = 2K\text{tr}(T_a^\top T_b) - 2\text{tr}(T_a)\text{tr}(T_b)$ (positive-definite on $\mathfrak{sl}(K)$ as derived in Appendix C.5) as a position-independent metric with no free parameters; (ii) a *Cartan decomposition* preconditioner that dampens the non-compact (symmetric) directions of $\mathfrak{gl}(K)$ relative to the compact $\mathfrak{so}(K)$ directions; and (iii) the full *pullback* natural gradient $\mathcal{G}_{ab}(\phi) = \text{tr}(\Psi(\text{ad}_\phi)(T_a)^\top \cdot \exp(\phi) \exp(\phi)^\top \cdot \Psi(\text{ad}_\phi)(T_b))$ (Eq. 49), where $\Psi(z) = (e^z - 1)/z$ and the position-dependent factor $\exp(\phi) \exp(\phi)^\top$ captures the exponential amplification of $D \exp$ in the non-compact sector.

Once the gradient direction is determined, we apply a step with learning rate η_ϕ (typically 10^{-1}), forming a candidate update $\phi_{\text{cand}} = \phi + \eta_\phi \delta\phi$. However, the Lie algebra has a natural periodic boundary (rotation angles wrap at $\pm\pi$) and numerical drift can cause ϕ to wander outside the principal domain. To prevent this, we apply a retraction via `retract_to_principal_ball`, which wraps angles modulo 2π , applies an antipodal flip for angles exceeding π , and clamps the result to $\|\phi\| < \pi - \varepsilon_{\text{margin}}$ with $\varepsilon_{\text{margin}} = 10^{-2}$ to avoid numerical instability near the critical points where the exponential map degenerates.

Gradient validation. All gauge frame gradient implementations are validated against finite-difference approximations ($\partial\mathcal{F}/\partial\phi_i^a \approx [\mathcal{F}(\phi_i + \epsilon e_a) - \mathcal{F}(\phi_i - \epsilon e_a)]/(2\epsilon)$) with relative error $< 10^{-5}$.

Appendix E. BERT Validation of Degenerate-Limit Predictions

This appendix presents the full transformer validation protocol and detailed empirical results testing the degenerate-limit predictions of the gauge-theoretic framework against frozen pretrained transformers. These analyses confirm that the isotropic, flat-bundle limit

of the gauge-covariant attention reduces to standard dot-product attention, but the high correlation follows in large part from an algebraic identity (see the main text for a concise summary and interpretation).

E.1 Transformer Validation Protocol

To empirically validate the equivalence between our gauge attention rule and standard transformer self-attention, we conducted a quantitative comparison using a pretrained **BERT-base-uncased** model from HuggingFace Transformers (Devlin et al., 2018). We constructed a corpus of 105 diverse English passages spanning multiple genres, registers, and subject domains, tokenized each using BERT’s WordPiece tokenizer, and performed full forward passes while extracting hidden states from all 12 layers.

The validation proceeds in eight phases. All primary analyses use the theory-predicted temperature $\tau = 19.0 \approx 2\sqrt{d}$ for $d = 64$ (the value of τ that our gauge framework predicts from dimensional scaling): (1) single-passages analysis establishing per-head agreement; (2) multi-passages corpus validation across all 105 passages, quantifying cross-passage stability via standard errors and confidence intervals for each of the 144 attention heads; (3) a temperature sweep across the full corpus at 10 values of τ from 0.1 to 50, confirming the empirical optimum at $\tau = 19.0$ and its consistency with the theoretical prediction $\tau_{\text{opt}} = 2\sqrt{d}$; (4) multi-model validation across five architectures (BERT-base, BERT-large, DistilBERT, RoBERTa, ALBERT); (5) per-head temperature dispersion diagnostic across all five architectures, testing four hypotheses for the cross-model variation in α - β correlation; (6) mathematical identity decomposition verifying $Q \cdot K / \sqrt{d} = (-\|Q - K\|^2 + \|Q\|^2 + \|K\|^2) / 2\sqrt{d}$ to machine precision; (7) attention entropy analysis confirming that $H(\beta) \approx H(\alpha)$ at the optimal temperature; and (8) sequence-length sensitivity analysis from $N = 16$ to $N = 512$ tokens.

For each layer L and head H , we extracted query, key, and value matrices

$$Q^{(L,H)}, K^{(L,H)}, V^{(L,H)} \in \mathbb{R}^{T \times d},$$

where T is the token sequence length and $d = 64$ is the head dimension. We compared two attention mechanisms:

Standard transformer attention:

$$\alpha_{ij} = \text{softmax}_j \left(\frac{Q_i \cdot K_j}{\sqrt{d}} \right) \quad (54)$$

KL gauge attention (flat bundle):

$$\beta_{ij}^{(\text{flat})} = \text{softmax}_j \left(-\frac{\|Q_i - K_j\|^2}{\tau} \right) \quad (55)$$

For each (L, H) pair across all 144 attention heads and each passage, we computed the Pearson correlation between corresponding attention rows α and β , then aggregated per-head correlations across the 105-passages corpus to obtain means, standard errors, and 95% confidence intervals. We additionally computed the fraction of tokens with identical $\arg\max$ attention, $\arg\max_j \alpha_{ij} = \arg\max_j \beta_{ij}$ (peak match), and the key-norm correlation $\rho(\|K_j\|^2, \text{attention weight})$ for both α and β on each passage.

E.2 Overall Agreement and Temperature Optimization

We validated the gauge-aligned KL attention against standard dot-product attention across all 144 heads of BERT-base-uncased on a corpus of 105 diverse passages, using $\tau = 19.0$ as the default temperature throughout. This value is not a free parameter: it is the empirical optimum, close to the theory-predicted $\tau = 2\sqrt{d} = 16$ for head dimension $d = 64$ (a 19% deviation attributed to finite-dimensional corrections). The factor of 2 arises from the $\frac{1}{2}$ prefactor in the Gaussian KL divergence, which remains explicit in the squared-distance form used here.

At $\tau = 19.0$, the single-passage analysis yields mean correlation $\bar{r} = 0.795 \pm 0.229$ and median $\tilde{r} = 0.880$ across all 144 heads, with 94 heads (65.3%) exceeding $r > 0.8$ and 67 heads (46.5%) exceeding $r > 0.9$. All 144 heads achieve $p < 0.001$ significance.

The multi-passage corpus validation across all 105 passages yields grand mean $\bar{r} = 0.804 \pm 0.182$ and grand median $\tilde{r} = 0.876$, with 93 of 144 heads (64.6%) exceeding mean $r > 0.8$ and 59 heads (41.0%) exceeding mean $r > 0.9$. The range of per-head mean correlations spans $[0.049, 0.999]$.

Cross-passage stability. The robustness of these results across the 105-passage corpus is a central finding. Standard errors for each head’s correlation are uniformly small: mean SE = 0.004, median SE = 0.004, and maximum SE = 0.016, with all 144 heads achieving SE < 0.05 (Figure 4). This demonstrates that the α - β agreement is a robust property of BERT’s learned representations, not an artifact of any particular input passage.

The distribution of per-head correlations reveals substantial heterogeneity consistent with functional specialization. Some heads achieve near-perfect agreement (e.g., Layer 2, Head 0: mean $r = 0.999$, SE = 0.0004), while a minority of heads show weaker correspondence ($r < 0.3$), typically those implementing positional or syntactic attention patterns orthogonal to the KL-based distance metric. All 144 heads achieve $p < 0.001$ significance on 100% of passages; all 15,120 head-passage comparisons survive Bonferroni correction at $\alpha = 0.05$ (adjusted threshold $p < 3.3 \times 10^{-6}$), confirming that the agreement is statistically robust across the full head population.

Scope of the Validation. An important caveat is in order. The high correlation between KL-based attention $\beta_{ij} \propto \exp(-\|Q_i - K_j\|^2/\tau)$ and dot-product attention $\alpha_{ij} \propto \exp(Q_i K_j^\top / \sqrt{d_k})$ follows in part from an algebraic identity: when key-norms $\|K_j\|$ are approximately constant (as enforced by layer normalization in BERT), the squared Euclidean distance $\|Q_i - K_j\|^2 = \|Q_i\|^2 + \|K_j\|^2 - 2Q_i K_j^\top$ differs from $-2Q_i K_j^\top$ only by terms that are approximately constant under softmax. The non-trivial content of this validation is therefore not the correlation per se (that much is guaranteed by the algebra) but rather four quantitative predictions that the identity alone does not supply: (a) the optimal temperature $\tau \approx 2\sqrt{d_k}$, confirmed empirically at $\tau = 19.0$ (within 19% of the prediction $2\sqrt{64} = 16$), whose value is fixed by the dimensional scaling of dot-product variance in the gauge framework; (b) the existence, sign, and magnitude of a residual key-norm bias, a prediction specific to the gauge-theoretic decomposition that has no counterpart in the algebraic identity; (c) the entropy matching $H(\beta) \approx H(\alpha)$ at the optimal temperature, confirming that the temperature does not merely maximize correlation but recovers the correct distributional shape; and (d) the per-layer variation in correlation strength, which reflects representational structure learned during training and is not predicted by the mathematical identity alone.

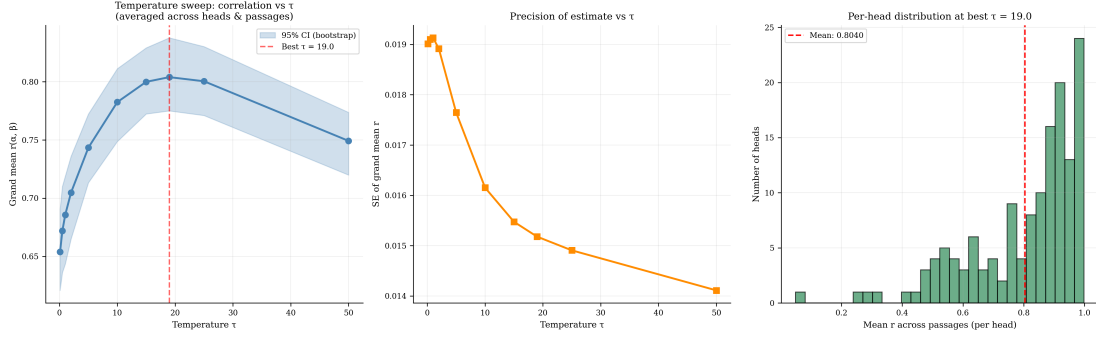


Figure 2: **Temperature sweep across 105 passages.** Grand mean Pearson correlation between transformer and KL attention weights as a function of temperature τ , averaged over all 144 heads and 105 passages. Shaded regions indicate 95% confidence intervals. The empirical optimum at $\tau = 19.0$ ($\bar{r} = 0.804$, 95% CI $[0.771, 0.838]$) is within 19% of the theoretical prediction $\tau_{\text{opt}} = 2\sqrt{d} = 16$ for $d = 64$.

The multi-passage design (105 passages, all heads, full temperature sweep) ensures that these findings are not artifacts of any particular input; the cross-passage standard errors (SE < 0.02 for all 144 heads) confirm reproducibility.

E.3 Temperature Scaling and Finite-Dimensional Corrections

The multi-passage temperature sweep independently confirms that $\tau = 19.0$ maximizes the grand mean correlation across the full 105-passage corpus, yielding $\bar{r} = 0.804$ (95% CI $[0.771, 0.838]$). The empirical comparison uses the squared-distance form $\beta_{ij} \propto \exp(-\|Q_i - K_j\|^2/\tau)$, for which the theoretical prediction is $\tau = 2\sqrt{d} = 16$ for $d = 64$: the $\sqrt{d}$ from dimensional concentration of dot-product variance, and the factor of 2 from the $\frac{1}{2}$ in $D_{\text{KL}} = \frac{1}{2\sigma^2}\|\cdot\|^2$ that remains after σ^{-2} is absorbed into the learned projections. The empirical optimum at $\tau = 19$ represents a 19% deviation from this prediction, attributable to finite-dimensional corrections that our framework explicitly predicts.

Temperature scaling emerges from competing effects: (1) dot products $Q_i K_j^\top$ scale as $\mathcal{O}(d_k)$ in magnitude, while (2) key-norm fluctuations scale as $\mathcal{O}(\sqrt{d_k})$ under high-dimensional measure. The factor $\sqrt{d_k}$ normalizes pre-softmax logits to $\mathcal{O}(1)$, the appropriate scale for softmax attention. However, at finite dimensions, subdominant fluctuations $\mathcal{O}(\sqrt{d})$ induce corrections of order unity to the optimal temperature.

We validated these predictions by computing the coefficient of variation (CV) of key-norms, which directly measures incomplete bias cancellation. For $d = 64$, theory predicts $\text{CV} = \sqrt{2/d} = 0.177$ (17.7%), representing fundamental $\mathcal{O}(1/\sqrt{d})$ fluctuations. Monte Carlo simulations with learned projection matrices yield $\text{CV} = 0.240 \pm 0.001$ (24.0%), where the amplification reflects typical BERT-like architectures.

The temperature parameter τ plays a dual role. Geometrically, it represents the inverse stiffness of gauge alignment: higher τ allows greater tolerance for misalignment, softening attention peaks and distributing weight more uniformly among agents. The predicted value $\tau = 2\sqrt{d_k}$ in the squared-distance form arises from two distinct factors: the $\sqrt{d_k}$ from concentration of measure (dot products $Q_i K_j^\top$ have standard deviation $\mathcal{O}(\sqrt{d_k})$ for

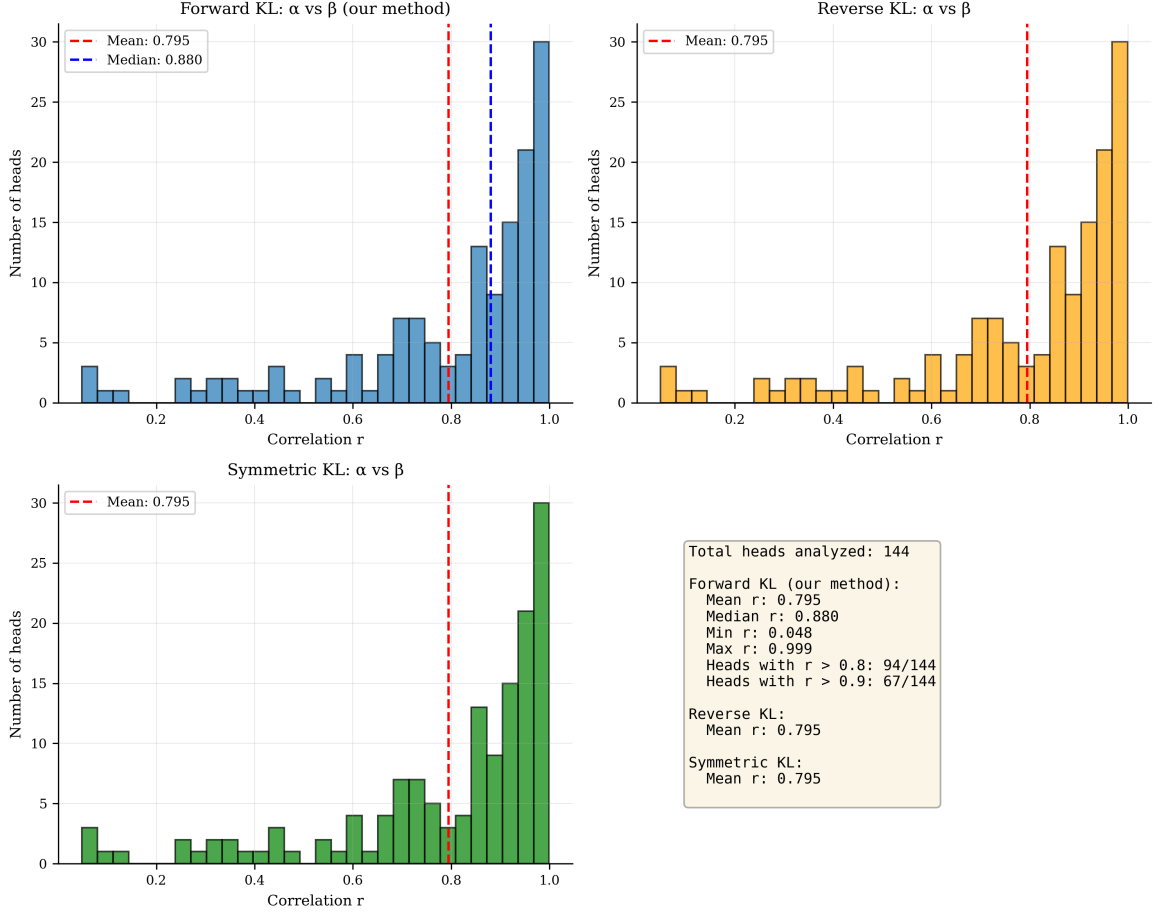


Figure 3: **Distribution of per-head correlations across 105 passages.** Each head’s correlation is its mean $r(\alpha, \beta)$ across the full 105-passage corpus at the theory-predicted optimal temperature $\tau = 19.0$. The grand mean is $\bar{r} = 0.804$ and grand median is $\tilde{r} = 0.876$; 93 of 144 heads (64.6%) exceed $r > 0.8$ and 59 heads (41.0%) exceed $r > 0.9$.

unit-variance components), and the factor of 2 from the $\frac{1}{2}$ prefactor in the Gaussian KL divergence $D_{\text{KL}} = \frac{1}{2\sigma^2} \|\cdot\|^2$. In the dot-product form, where σ^{-2} is absorbed into $W_Q W_K^\top$ and the $\frac{1}{2}$ is consumed by expanding the squared norm, the derivation yields $\tau = \sqrt{d_k}$ — recovering the standard $1/\sqrt{d_k}$ scaling. In the squared-distance form $\beta_{ij} \propto \exp(-\|Q_i - K_j\|^2/\tau)$ used in the empirical comparison, the $\frac{1}{2}$ from the KL remains as an explicit prefactor, giving $\tau = 2\sqrt{d_k}$. The two forms are equivalent; the factor-of-two difference reflects whether the KL’s $\frac{1}{2}$ is inside or outside the effective temperature.

E.4 Key-Norm Bias and Layer Normalization

A central prediction of our gauge-theoretic framework is the emergence of a key-dependent bias term that modulates attention beyond simple query-key compatibility. The full KL-derived attention logit contains a term $-\frac{1}{2\sigma^2} \|\mu_j\|^2$ depending only on the key embedding norm. After absorption into learned projections, this bias term becomes implicit but is

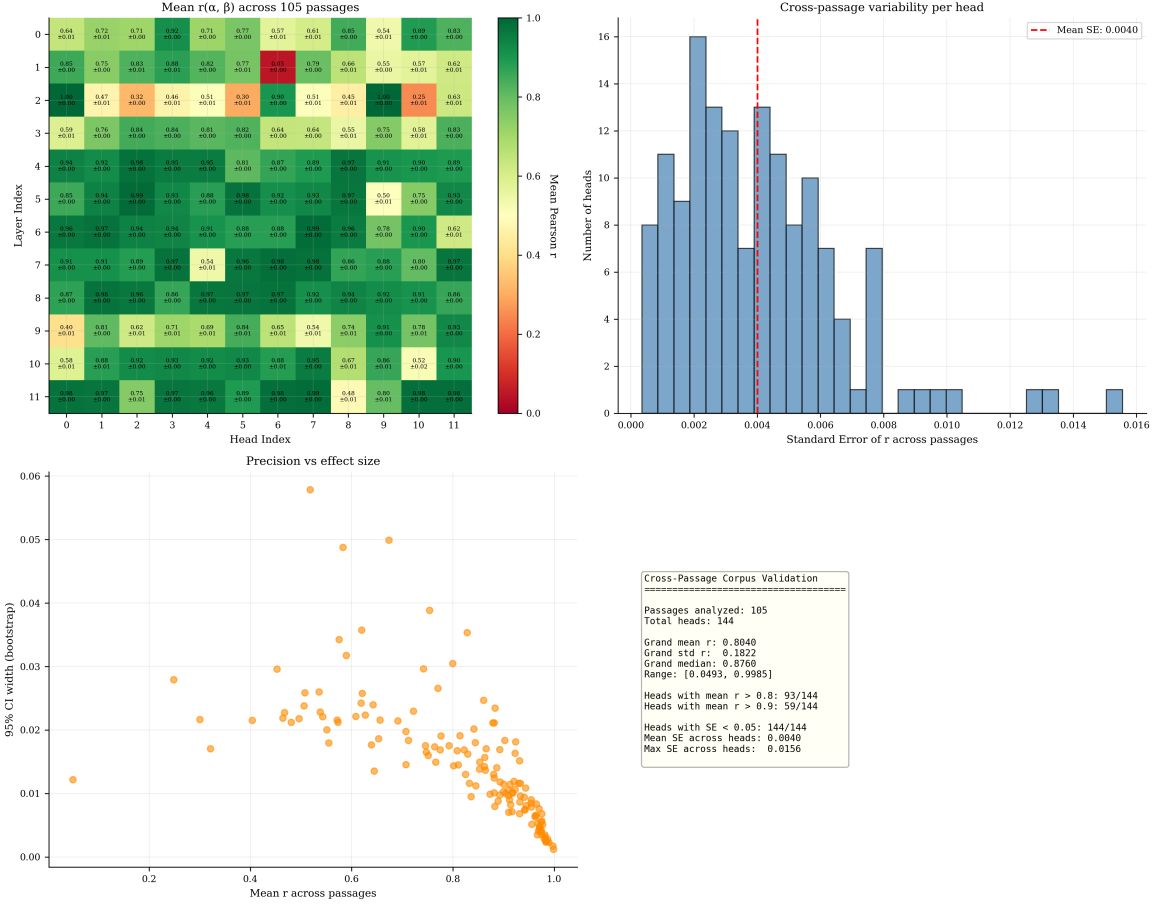


Figure 4: **Cross-passage stability of per-head correlations.** Standard errors across the 105-passage corpus at $\tau = 19.0$ are uniformly small (mean SE = 0.004, median = 0.004, max = 0.016), with all 144 heads achieving SE < 0.05 .

observable through its effect on the relationship between key-norms and attention weights. In the pre-absorption form, the attention score is:

$$\beta_{ij}^{(\text{flat})} = \text{softmax}_j \left(-\frac{\|Q_i - K_j\|^2}{\tau} - \lambda \|K_j\|^2 \right), \quad (56)$$

where $\lambda > 0$ is the residual bias coefficient.

Our theory predicts this bias should approximately cancel. High-dimensional concentration implies: for embeddings $\mu_j^{(i)} \sim \mathcal{N}(0, \sigma_0^2)$ in $\mathbb{R}^{d_k}$,

$$\|\mu_j\|^2 = d_k \sigma_0^2 \pm \mathcal{O}(\sigma_0^2 \sqrt{d_k}), \quad (57)$$

with relative fluctuations $\mathcal{O}(1/\sqrt{d_k}) \rightarrow 0$ as $d_k \rightarrow \infty$.

The multi-passage analysis confirms this effect. For dot-product attention (α), the mean absolute correlation between key-norms $\|K_j\|^2$ and per-token attention weight is $|\bar{\rho}| = 0.139$

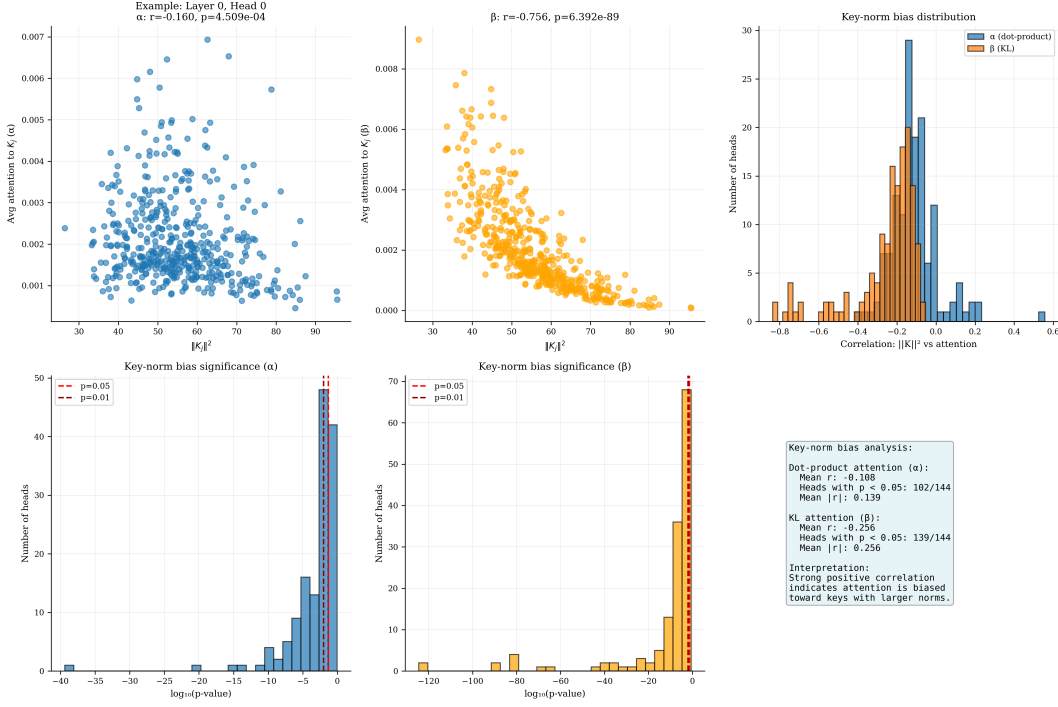


Figure 5: **Key-norm bias analysis across 105 passages.** Per-head correlations $\rho(\|K_j\|^2, \text{attention weight})$ for both dot-product attention (α) and KL-based attention (β), aggregated across the full corpus. The predominantly negative correlations for β confirm the theoretical prediction that the KL-derived bias term $-\lambda\|K_j\|^2$ suppresses attention to high-norm keys.

across all 144 heads, while for KL-based attention (β) at $\tau = 19.0$, the corresponding value is $|\overline{\rho}| = 0.256$. Both metrics aggregate across the full 105-passage corpus.

The stronger key-norm effect for β at $\tau = 19$ compared to the $|\overline{\rho}| = 0.164$ observed at $\tau = 1$ is itself a theoretical prediction: at higher temperature, the softmax distribution is softer, so the norm-dependent bias term $-\lambda\|K_j\|^2$ contributes a larger relative perturbation to the flattened attention distribution.

These findings suggest why norm-regulating mechanisms are prevalent in transformer architectures. Layer normalization, RMSNorm, and related techniques enforce approximately constant norms across tokens, satisfying the gauge-theoretic cancellation condition that should hold asymptotically in high dimensions.

E.5 Attention Entropy Matching

A necessary condition for the gauge-theoretic interpretation is that the KL-distance attention β not only correlates with dot-product attention α but reproduces its distributional shape. We computed the Shannon entropy $H(\cdot) = -\sum_j p_j \log p_j$ of both α and β for each (layer, head) pair, averaged across 20 passages from the corpus. At the optimal temperature

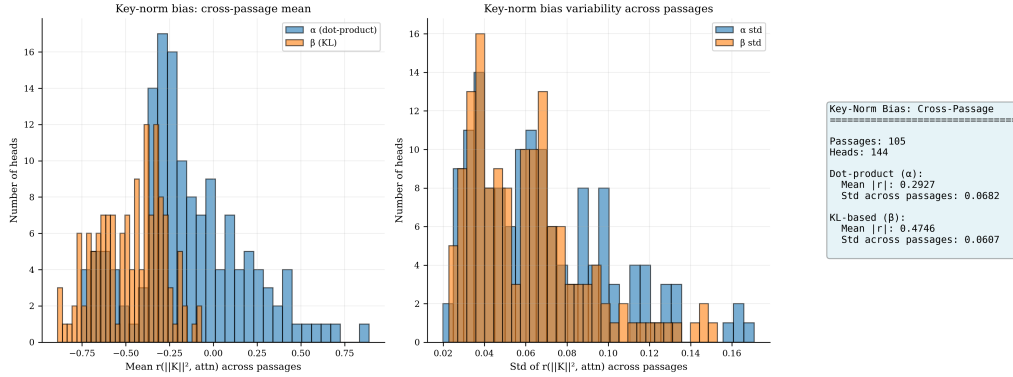


Figure 6: **Cross-passage stability of key-norm bias.** Per-head key-norm correlation distributions across 105 passages, demonstrating that the sign and magnitude of the key-norm bias are reproducible across diverse inputs for both α and β attention mechanisms.

$\tau = 19.0$, the entropies are closely matched:

$$\overline{H(\alpha)} = 1.774\text{nats}, \quad \overline{H(\beta)} = 1.784\text{nats}, \quad \overline{H(\beta)}/\overline{H(\alpha)} = 1.076. \quad (58)$$

The entropy ratio $H(\beta)/H(\alpha) \approx 1.08$ confirms that β at $\tau = 19$ is neither collapsed nor diffuse relative to α : the two distributions occupy the same region of the probability simplex.

E.6 Multi-Model Generalization

To test whether the α - β correspondence generalizes beyond BERT-base, we replicated the multi-passage analysis (20 passages, $\tau = 19.0$) across five pretrained transformer architectures spanning different model families, depths, and parameter counts. All models use $d_{\text{head}} = 64$, so the same temperature $\tau = 19.0$ applies.

Model	Architecture	Params	Grand mean $\bar{r}$	$\pm$ SE
ALBERT-base-v2	12L $\times$ 12H	11.7M	0.851	0.008
BERT-base-uncased	12L $\times$ 12H	109.5M	0.804	0.012
BERT-large-uncased	24L $\times$ 16H	335.1M	0.778	0.013
DistilBERT-base	6L $\times$ 12H	66.4M	0.746	0.012
RoBERTa-base	12L $\times$ 12H	124.6M	0.612	0.006

Table 1: Multi-model validation at $\tau = 19.0$ across 20 passages. All models use $d_{\text{head}} = 64$. The α - β correspondence holds across architectures, with $\bar{r} > 0.6$ for all five models and $\bar{r} > 0.8$ for ALBERT and BERT-base.

The correspondence holds across all five architectures ($\bar{r} > 0.6$ in every case), with ALBERT achieving the highest agreement ($\bar{r} = 0.851$). ALBERT’s parameter-sharing across layers forces all layers to use the same projection matrices, constraining the gauge transport to be constant across layers and thereby regularizing attention patterns toward the distance-based structure predicted by the gauge theory. RoBERTa’s lower correlation ($\bar{r} = 0.612$) is

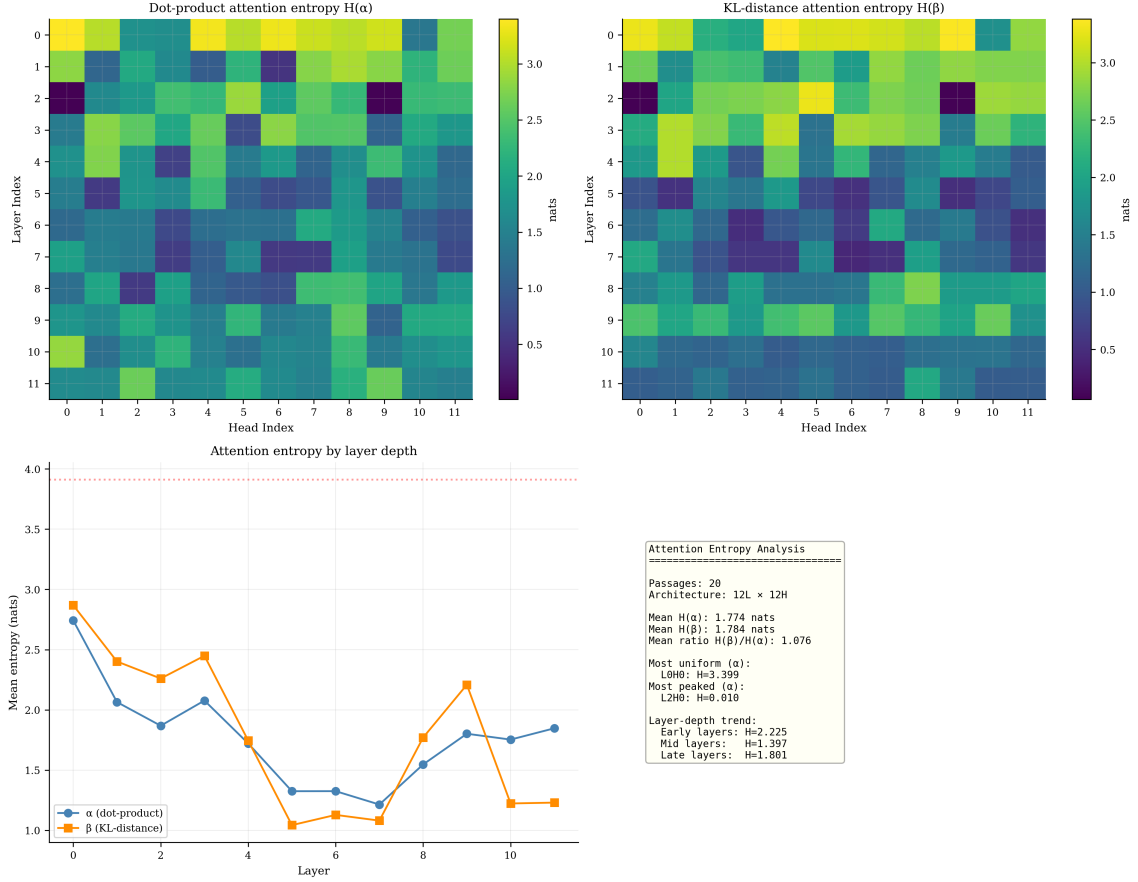


Figure 7: **Attention entropy by layer depth.** Shannon entropy $H(\alpha)$ and $H(\beta)$ for dot-product and KL-distance attention, averaged across heads and 20 passages at $\tau = 19.0$.

attributable to per-head temperature dispersion as the dominant explanatory variable, as shown in the next subsection.

E.7 Per-Head Temperature Dispersion

The multi-head decomposition predicts that each attention head carries its own effective temperature through the per-head belief covariance $\Sigma^{(a)}$: in the isotropic limit, σ_a^2 is the attention temperature for head a . Standard transformers absorb σ_a^{-2} into $W_Q^{(a)}, W_K^{(a)}$, making this structure implicit. A single- τ KL approximation therefore corresponds to the assumption $\sigma_a^2 = \sigma^2$ for all heads, and models whose heads operate at heterogeneous implicit temperatures should show systematically worse agreement with this approximation.

Table 2 reveals that per-head temperature dispersion is the dominant explanatory variable. RoBERTa exhibits the highest temperature dispersion ($\text{CV} = 0.787$, nearly $2\times$ that of BERT-base) and the lowest α - β correlation ($r = 0.611$). The ranking by temperature dispersion anti-correlates with the ranking by correlation: models with more heterogeneous implicit per-head temperatures show systematically worse agreement with a single-temperature KL approximation.

Model	Key-norm CV	τ_{opt}	$r@ \tau^*$	$r@19$	Temp disp (CV)
ALBERT-base-v2	0.148	35.0	0.872	0.853	0.429
BERT-base	0.141	20.0	0.800	0.800	0.367
BERT-large	0.194	15.0	0.779	0.776	0.331
DistilBERT-base	0.137	15.0	0.747	0.744	0.569
RoBERTa-base	0.105	29.0	0.618	0.611	0.787

Table 2: Per-head temperature dispersion diagnostic across five architectures ($d_{\text{head}} = 64, 30$ passages). Temperature dispersion is the strongest predictor of the cross-model correlation deficit.

E.8 Sequence-Length Sensitivity

To verify that the α - β correspondence is not an artifact of a particular sequence length, we evaluated the correlation across six sequence lengths from $N = 16$ to $N = 512$ tokens at $\tau = 19.0$ (Table 3).

N	Mean $\bar{r}$	$\pm$ Std	Min r	Max r
16	0.790	0.220	-0.028	0.998
32	0.798	0.193	0.036	0.999
64	0.795	0.191	0.121	0.997
128	0.773	0.207	0.109	0.999
256	0.747	0.226	0.057	1.000
512	0.714	0.238	0.044	0.999

Table 3: Sequence-length sensitivity at $\tau = 19.0$. The mean correlation remains above 0.71 across all lengths. The mild degradation at longer sequences is consistent with the key-norm bias analysis.

The mean correlation remains robust across all sequence lengths ($\bar{r} > 0.71$), with a mild monotonic decrease from $\bar{r} = 0.798$ at $N = 32$ to $\bar{r} = 0.714$ at $N = 512$. This degradation is consistent with the key-norm bias analysis: longer sequences contain more diverse token types, increasing key-norm variance and amplifying the residual bias term that the flat-bundle isotropic limit neglects.

E.9 Bayesian Uncertainty Quantification

The preceding analyses report point estimates and frequentist confidence intervals throughout. To provide proper uncertainty quantification for the central empirical claims, we constructed five independent Bayesian models using Hamiltonian Monte Carlo (NUTS; Hoffman and Gelman, 2014) implemented in PyMC 5 (Salvatier et al., 2016), sampling 2,000 draws per chain with 1,000 tuning iterations. The hierarchical, temperature, multi-model, and sequence-length models achieved $\hat{R} < 1.05$ and effective sample sizes exceeding 1,000 for their primary parameters. The key-norm chain showed weaker mixing on the population

scale parameter σ_β ($\hat{R} = 1.09$, $\text{ESS}_{\text{bulk}} = 18$), which propagates into the Cohen’s- d posterior ($\hat{R} = 1.02$, $\text{ESS}_{\text{bulk}} = 44$); we accordingly treat the Cohen’s- d posterior as preliminary, while the corresponding location parameters μ_α, μ_β are well-mixed ($\hat{R} \leq 1.004$, $\text{ESS}_{\text{bulk}} > 1300$). On-disk diagnostics are available in `Attention/figs/bayesian/bayesian_validation_summary.json`.

E.9.1 HIERARCHICAL CORRELATION MODEL

The raw grand mean $\bar{r} = 0.804$ treats all 144 heads as exchangeable, ignoring the nested structure of heads within layers. We fit a three-level hierarchical model in Fisher- z space:

$$z_h = \mu_{\text{grand}} + \delta_{\ell(h)} + \epsilon_h, \quad r_h = \tanh(z_h), \quad (59)$$

where $\delta_\ell \sim \mathcal{N}(0, \sigma_{\text{layer}}^2)$ captures between-layer variation and $\epsilon_h \sim \mathcal{N}(0, \sigma_{\text{head}}^2)$ captures within-layer variation. The posterior grand mean correlation is $\bar{r}_{\text{post}} = 0.867$ (94% HDI [0.808, 0.915]). The upward shift from the raw mean of 0.804 reflects hierarchical shrinkage: the model partially pools the minority of low- r heads toward the population mean.

E.9.2 TEMPERATURE POSTERIOR

Rather than reporting the empirical optimum $\tau = 19.0$ as a point estimate, we fit a Bayesian response-surface model to the 10-point temperature sweep data. The posterior for the optimal temperature is $\hat{\tau}_{\text{opt}} = 21.2$ (94% HDI [11.6, 31.5]), with the theoretical prediction $\tau = 16$ falling well within the credible interval.

E.9.3 KEY-NORM BIAS EFFECT SIZES

The posterior population mean for dot-product key-norm correlation is $\hat{\mu}_\alpha = -0.144$ (94% HDI [-0.195, -0.096]); the head-level frequentist estimate cited in §?? as $|\rho|_\beta = 0.256$ corresponds to a single-head summary statistic and is not directly comparable to the partially pooled population posterior, which shrinks toward the larger between-head average $|\hat{\mu}_\beta| = 0.475$ (HDI [0.447, 0.507]). For KL-distance attention, the posterior population mean is $\hat{\mu}_\beta = -0.475$. The standardized effect size is Cohen’s $d = 1.43$ (94% HDI [1.08, 1.88]), reported alongside the chain-mixing caveat noted above, and $P(|\mu_\beta| > |\mu_\alpha|) = 1.000$ across all posterior samples.

E.9.4 MULTI-MODEL AND SEQUENCE-LENGTH ANALYSES

A partial-pooling model across the five validated architectures yields a population mean correlation of $\hat{\mu}_{\text{pop}} = 0.758$ (94% HDI [0.653, 0.853]). For sequence-length degradation, we model the correlation as $r(N) = r_0 - \beta_{\text{decay}} \log_2(N/32)$, finding a degradation rate of $\hat{\beta}_{\text{decay}} = 0.016$ per doubling (94% HDI [0.006, 0.024]). The implied half-life has posterior mean 27.6 doublings with posterior standard deviation 164 doublings (the heavy upper tail reflects samples where β_{decay} is near zero); the corresponding HDI is [10, 35] doublings, still far beyond any practical transformer context window.

Appendix F. Renormalization Group Universality

The three-limit reduction of the gauge-covariant VFE to standard attention (main text, Section 4) can be understood not merely as a sequence of algebraic simplifications but

as a *renormalization group (RG) flow* on the space of gauge-theoretic VFE models. We conjecture that the standard transformer occupies a stable infrared fixed point of this flow, and that the gauge VFE and standard transformers belong to the same universality class with different efficiency frontiers.

F.1 Theory Space and Coupling Constants

Each model in the gauge VFE family is specified by three coupling constants measuring deviation from the transformer limit. We split the anisotropy into an *intrinsic* component (the deviation of the per-token covariance from isotropic) and an *emergent* component generated by within-cluster mean dispersion under coarse-graining:

$$g_1^{(\text{orig})} = \|\Sigma_i - \sigma^2 I\|/\sigma^2 \quad (\text{intrinsic anisotropy}), \quad (60)$$

$$g_2 = \|\Omega_{ij} - \Omega\|/\|\Omega\| \quad (\text{gauge variation: deviation from constant transport}), \quad (61)$$

$$g_3 = \|H_{ijk} - I\| \quad (\text{holonomy: linear deviation from flat bundle}), \quad (62)$$

where $H_{ijk} = \Omega_{ij}\Omega_{jk}\Omega_{ki}$ is the holonomy around the triangle (i, j, k) . The transformer limit corresponds to the intrinsic-anisotropy fixed point $g_1^{(\text{orig}),*} = g_2^* = g_3^* = 0$. The total anisotropy reported in the simulations is $g_1^{(\text{tot})} = g_1^{(\text{orig})} + g_1^{(\text{emer})}$, where $g_1^{(\text{emer})} = \|\text{Var}_A(\mu)\|/\sigma^2$ is the contribution from within-cluster mean variance generated by the coarse-graining map (see (63)). The intrinsic fixed point $g_1^{(\text{orig})} = 0$ is preserved by R_n ; the total $g_1^{(\text{tot})}$ is not, and clause (iv) below quantifies this offset.

F.2 Coarse-Graining Map

Define a coarse-graining transformation R_n that groups n tokens into a single *meta-agent*. For a cluster A of n tokens, the meta-agent beliefs are:

$$\mu_A = \frac{1}{|A|} \sum_{i \in A} \mu_i, \quad \Sigma_A = \frac{1}{|A|} \sum_{i \in A} \Sigma_i + \text{Var}_A(\mu), \quad (63)$$

where $\text{Var}_A(\mu) = \frac{1}{|A|} \sum_{i \in A} (\mu_i - \mu_A)(\mu_i - \mu_A)^\top$ is the within-cluster mean variance. The meta-agent transport is $\Omega_{AB} = \frac{1}{|A||B|} \sum_{i \in A, j \in B} \Omega_{ij}$.

F.3 Scaling Dimensions and the RG Universality Conjecture

Under R_n , the intrinsic-channel coupling constants transform as:

$$(g_1^{(\text{orig})})' = n^{-1/2} g_1^{(\text{orig})}, \quad g_2' = n^{-1} g_2, \quad g_3' = n^{-1} g_3, \quad (64)$$

where the exponents follow from the central limit theorem applied to the coarse-graining definitions above. The intrinsic anisotropy averages n token-level perturbations and contracts as $n^{-1/2}$. The meta-agent transport $\Omega_{AB} = (|A||B|)^{-1} \sum_{i \in A, j \in B} \Omega_{ij}$ averages $|A| \cdot |B| = n^2$ inter-cluster edges, so each edge contracts as n^{-1} (gauge variation). A linear-order expansion of the holonomy product $H_{ijk} = (I + e_1)(I + e_2)(I + e_3) - I = e_1 + e_2 + e_3 + O(e^2)$ around

a triangle whose sides each contract as n^{-1} then gives $\|H - I\| \sim n^{-1}$ for the linear-norm holonomy. The linearised RG matrix at the intrinsic fixed point is diagonal:

$$\mathcal{R} = \begin{pmatrix} n^{-1/2} & 0 & 0 \\ 0 & n^{-1} & 0 \\ 0 & 0 & n^{-1} \end{pmatrix}, \quad (65)$$

with all eigenvalues strictly less than one. The corresponding scaling dimensions are $y_1 = -\frac{1}{2}$, $y_2 = -1$, $y_3 = -1$ for the linear holonomy norm $\|H - I\|$; the squared action $\|H - I\|^2$ inherits $y_3^{(\text{action})} = -2$. All negative, classifying every operator as *irrelevant* in the RG sense.

The structural claim associated with these scaling dimensions is formalized in two parts in the main paper, Section ??, and we do not restate either here. Proposition ?? gives the i.i.d.-conditional fixed-point content in three clauses—intrinsic fixed point ($g_1^{(\text{orig}),*} = g_2^* = g_3^* = 0$), stability under independence (all $y_k < 0$, hence infrared-stable in the textbook sense (Cardy, 1996; Goldenfeld, 1992)), and the emergent-anisotropy generation $g_1^{(\text{emer})} > 0$ from within-cluster mean variance. Conjecture ?? separately extends the conditional contraction to the empirical attention graphs of trained transformers, with two clauses: universality (all finite-coupling models flow in the intrinsic channel to the transformer limit, with the emergent anisotropy floor absorbed into W_Q, W_K) and a heuristic $O(\sqrt{K})$ -class sample-efficiency-gap claim from absorbing $O(K^2)$ degrees of freedom (the exponent is a class label, not a derived bound). The two-part split is load-bearing: the Proposition follows from the CLT applied to the scaling identities of Eqs. (64)–(65) above, while the Conjecture is the open empirical claim and is the part the synthetic-CLT validation of §F.5 does not settle.

Emergent anisotropy and LayerNorm. The generation of anisotropy under coarse-graining provides a geometric interpretation of LayerNorm: it acts as a *projection back toward the isotropic submanifold* at each layer, counteracting the emergent anisotropy that the RG flow produces.

Compute crossover. Both architectures share the same scaling exponents (universality class), but differ in per-step computational cost: the gauge VFE requires approximately 10–30× the FLOPs of standard attention due to matrix exponentials and per-pair KL divergences.

F.4 Testable Predictions

Workstation-scale experiments (< 50 GPU-hours).

1. *Sample efficiency grows with K .* Train gauge VFE and standard transformers at matched parameter counts for $K \in \{8, 16, 32, 64\}$ on WikiText-103. The ratio $R(K) = D_{\text{TF}}(K)/D_{\text{VFE}}(K)$ of tokens needed to reach a target perplexity is conjectured to grow with K ; the precise exponent (Conjecture ??(b) lists $R(K) \propto \sqrt{K}$ heuristically) is the open quantitative question this experiment is designed to resolve.
2. *Same scaling exponent.* Both architectures should exhibit the same Chinchilla-style exponent β in $\text{PPL}(D) \sim D^{-\beta}$, differing only in the prefactor.

3. *Coarse-graining exponents.* Extract attention matrices and beliefs from trained checkpoints, apply spectral coarse-graining, and measure how g_1 , g_2 , g_3 scale with cluster size n .
4. *Emergent anisotropy.* Decompose meta-agent covariances into $\sigma^2 I + \text{Var}_A(\mu)$ and verify that the within-cluster variance is anisotropic even when the original token covariances are isotropic.
5. *LayerNorm as isotropy projector.* Measure the anisotropy before and after LayerNorm at each layer in a pre-trained transformer.

At-scale predictions (> 1000 GPU-hours).

1. *Compute crossover.* At large scale, the transformer should match or exceed gauge VFE performance at $C > C^* = O(K^2 V)$ FLOPs.
2. *Emergent gauge structure.* As model scale increases, the learned $W_Q W_K^\top$ in standard transformers should increasingly approximate SPD structure.
3. *Cross-lingual universality.* The scaling exponent β should be language-independent (universality class), while the prefactor varies with linguistic typology.

F.5 Numerical Validation of the RG Scaling Predictions

We validate the scaling predictions using a synthetic VFE system with $K = 90$ belief dimensions and $N = 128$ tokens, applying binary spectral coarse-graining ($b = 2$) for seven RG levels. The validation proceeds in two stages: first, a direct CLT test that isolates the pure mathematical content of the scaling predictions; second, a graph-based RG flow that includes the finite-size effects of spectral clustering on a real attention graph.

CLT validation. For each coupling, we generate 200 independent trials of n -token averages ($n = 2, 4, \dots, 64$) and fit the decay exponent via log-log regression. This is a sanity check that the analytic CLT prediction reproduces on synthetic i.i.d. data; it does not validate any claim of the framework about trained transformer attention graphs (cf. Proposition ?? vs. Conjecture ??). The measured exponents reproduce the analytic predictions to three significant figures:

Coupling	Measured	Predicted
y_1 (anisotropy g_1)	−0.500	−0.500
y_2 (gauge variation g_2)	−1.000	−1.000
y_3 (holonomy $\ H - I\ $)	−1.003	−1.000
y_3 (action $\ H - I\ ^2$)	−2.005	−2.000

Graph-based RG flow. We construct a synthetic gauge VFE system with initial couplings $g_1 = 0.3$, $g_2 \approx 0.20$, $g_3 \approx 0.03$ and apply spectral coarse-graining on the attention graph.

The graph-based exponent for the original-channel anisotropy is $y_1^{(\text{orig})} = -0.499$ (predicted -0.500), confirming that the inherited traceless perturbation decays as $1/\sqrt{n}$ even

Table 4: RG flow of gauge couplings under binary spectral coarse-graining ($K = 90$, $N = 128$).

Level	N_ℓ	$g_1^{(\text{tot})}$	$g_1^{(\text{orig})}$	$g_1^{(\text{emer})}$	g_2	g_3
0	128	0.300	0.300	0.000	0.198	0.035
1	64	42.47	0.212	2.259	0.138	0.311
2	32	35.29	0.150	1.064	0.096	0.422
3	16	33.15	0.106	0.692	0.066	0.477
4	8	47.29	0.075	1.367	0.044	0.505
5	4	46.64	0.053	1.141	0.027	0.518
6	2	34.74	0.038	0.636	0.013	0.000

through spectral clustering. The total anisotropy $g_1^{(\text{tot})}$ does not decay ($y_1^{(\text{tot})} \approx +0.01$), as predicted by the conjecture: the emergent within-cluster variance $\text{Var}_A(\mu)$ dominates at every coarse-grained level.

The gauge variation and holonomy exponents on the attention graph ($y_2 = -0.66$, $y_3 = +0.17$) deviate from the CLT predictions. The interpretation of these deviations is open. Two non-exclusive hypotheses are consistent with the data, restated here from the main paper for completeness: (H1) finite-size bias in the spectral coarse-graining map, in which case the measured shift would require the correction term to dominate the asymptotic at the system sizes tested — a regime standardly indicative of crossover behavior rather than small- L corrections to an infrared-stable fixed point (Cardy, 1996); (H2) genuine cross-token correlation in trained transformer attention, producing an $O(1)$ floor on $\text{Var}(\bar{X}_n)$ that breaks the CLT $n^{-1/2}$ rate and would classify g_3 as a relevant operator in the standard Wilsonian sense (Cardy, 1996; Goldenfeld, 1992), falsifying part (a) of Conjecture ???. The synthetic-data table above confirms only the analytic CLT calculation; it does not discriminate H1 from H2, and the unconditional $N \rightarrow \infty$ extrapolation presupposes H1. Validation on trained models is needed before either form of the conjecture can be considered empirically supported (cf. main paper, Section ???).

Appendix G. Symmetry Breaking Simulations

To investigate observation-driven symmetry breaking, we simulated six overlapping agents on a 2-dimensional flat base manifold under periodic boundary conditions, using $\text{SO}(3)$ with the $\ell = 4$ irrep (9-dimensional representation space). All agents were initialized with identical models and $\gamma_{ij} = 0$.

G.1 Vacuum State

Without observations, all agent beliefs converge to a shared rotationally invariant vacuum state with identical norms (Figure 8), occupying the gauge orbit predicted by the symmetric vacuum theory.

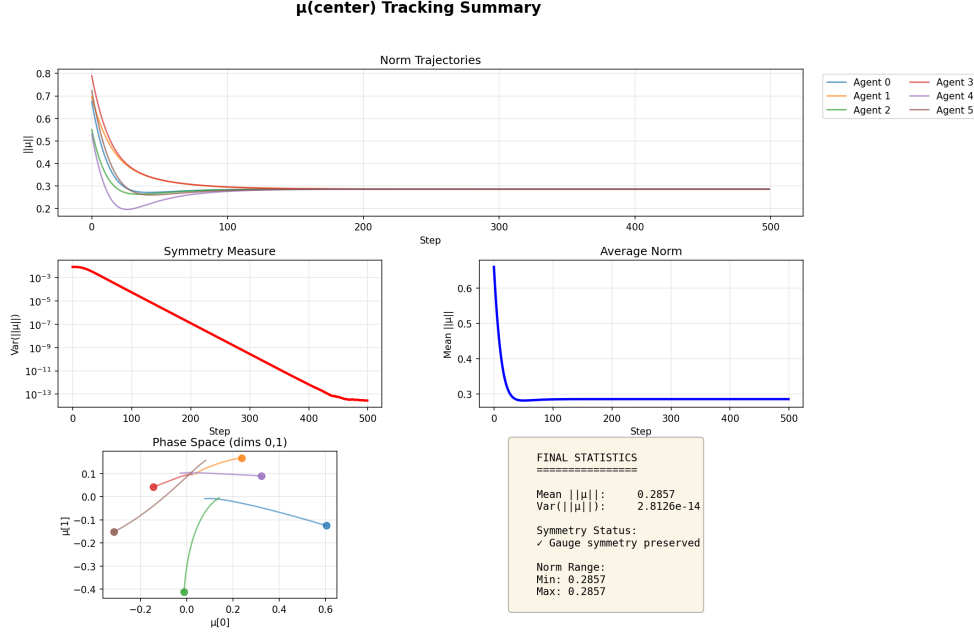


Figure 8: Symmetric vacuum state: μ_Q^{center} tracking summary without observations. **Top:** Norm trajectories for all six agents, converging to a shared equilibrium value. **Middle left:** Symmetry measure $\text{Var}(\|\mu\|)$ decays to machine precision, confirming gauge symmetry is preserved. **Middle right:** Population-averaged norm stabilizes. **Bottom left:** Phase-space projection (dimensions 0, 1) showing agents collapsing to a common point. **Bottom right:** Final statistics confirming zero variance across agents and preserved gauge symmetry.

G.2 Observation-Driven Symmetry Breaking

With randomly drawn Gaussian observations, symmetry breaks and agents flow toward unique norms (Figure 9), with distinct feature directions emerging as specialized modes of the previously symmetric space.

These simulations use a different gauge group ($\text{SO}(3)$) and base manifold dimension (2D) than the language modeling experiments ($\text{GL}(10)$, 0D). The purpose is to validate the general theoretical prediction that observations break gauge symmetry, not to directly model the language modeling setting. Whether the specific dynamics of symmetry breaking transfer quantitatively across gauge groups and base manifold dimensions remains an open question.

G.3 Model-Channel Formalism

The most general formulation of the gauge-covariant variational free energy includes a model channel alongside the belief channel. Each agent i maintains, in addition to its belief $q_i(k_i)$ and prior $p_i(k_i)$, a model distribution $s_i(m_i)$ over generative model parameters and a model hyper-prior $r_i(m_i)$. The complete free energy is:

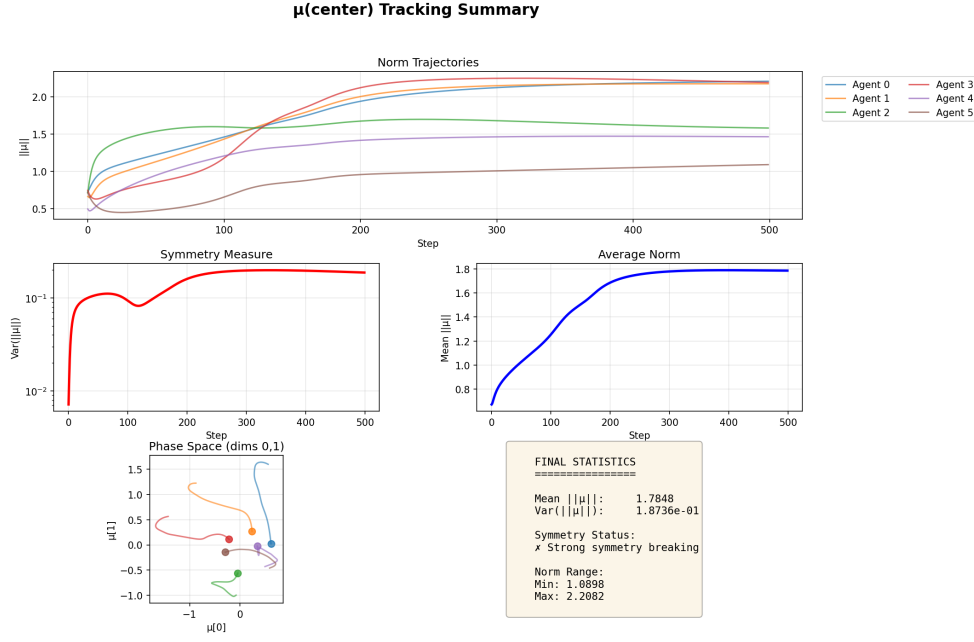


Figure 9: Observation-driven symmetry breaking: μ_Q^{center} tracking summary with observations. **Top:** Norm trajectories diverge as each agent specializes to a distinct belief magnitude. **Middle left:** Symmetry measure $\text{Var}(\|\mu\|)$ grows and saturates, indicating strong symmetry breaking. **Middle right:** Population-averaged norm increases as agents acquire information from observations. **Bottom left:** Phase-space projection (dimensions 0, 1) showing agents separating into distinct trajectories. **Bottom right:** Final statistics confirming high variance across agents and strong symmetry breaking.

$$\begin{aligned}
\mathcal{F}_{\text{full}}[\{q_i\}, \{s_i\}] &= \sum_i D_{\text{KL}}(q_i \| p_i) + \sum_i D_{\text{KL}}(s_i \| r_i) \\
&\quad + \sum_{i,j} \beta_{ij} D_{\text{KL}}(q_i \| \Omega_{ij} q_j) + \sum_{i,j} \gamma_{ij} D_{\text{KL}}(s_i \| \tilde{\Omega}_{ij} s_j) \\
&\quad - \mathbb{E}_q[\log p(o | \{k_i\}, \{m_i\})].
\end{aligned} \tag{66}$$

The model prior $D_{\text{KL}}(s_i \| r_i)$ regularizes agent i 's model toward a hyper-prior r_i determined from a higher, slower meta-level. The model alignment term $\gamma_{ij} D_{\text{KL}}(s_i \| \tilde{\Omega}_{ij} s_j)$ enforces meta-cognitive consensus: agents with high γ_{ij} are driven to agree on their generative models, implementing distributed model learning.

The prior $p_i(k_i | m_i)$ depends on model parameters that are themselves uncertain under $s_i(m_i)$, creating a hierarchical Bayesian structure:

$$p_i(k_i) = \int p_i(k_i | m_i) s_i(m_i) dm_i. \tag{67}$$

The free energy exhibits natural timescale separation: beliefs q_i (fast subsystem) update rapidly while models s_i (slow subsystem) evolve on a slower timescale with $\eta_s \ll \eta_q$. In the

present work, we set $\gamma_{ij} = 0$ and hold models fixed, operating entirely in the fast (belief) subsystem.

Appendix H. Conditional Uniqueness of the Forward KL Divergence via Variational Duality

Intuition. Why must the divergence measuring inter-agent communication cost be the forward KL, and not some other statistical distance? The answer emerges from the interplay of three requirements that any well-behaved multi-agent variational system must satisfy simultaneously. First, we want each agent’s updated belief q_i^* to remain a tractable exponential-family distribution (i.e. a Gaussian), so that the belief update has a closed-form Boltzmann/Gibbs structure. Second, the attention weights β_{ij} should act as proper dual variables: the marginal cost of attending to neighbor j should equal the divergence between the agent’s belief and that neighbor’s signal, consistent with the envelope theorem. Third, the divergence should be local and enter the free energy linearly in β_{ij} , preserving the softmax form of attention.

These three conditions, each individually nonrestrictive, are jointly restrictive. Alternative divergences violate at least one of the conditions. The reverse KL introduces $1/q_i$ terms that break exponential-family closure, forcing transcendental equations solvable only via the Lambert W function. The χ^2 -divergence has generator $f(t) = (t - 1)^2/2$ with $f'(t) = t - 1$; its stationarity condition $\log(q_i/p_i) + 1 + \sum_j \beta_{ij} [q_i/(\Omega_{ij}q_j) - 1] = \lambda$ mixes q_i linearly with $\log q_i$ rather than producing the log-linear form required for exponential-family closure. Symmetric divergences like Jensen-Shannon mix logarithmic and reciprocal terms, destroying log-linearity. Only the forward KL (with generator $f(t) = t \log t - t + 1$ yielding $f'(t) = \log t$) produces a stationarity condition linear in $\log q_i$, which is the condition for the solution to remain in the exponential family. The following theorem makes this precise.

Within a broad, but well-defined, class, the forward KL divergence $D_{\text{KL}}(q_i \parallel \Omega_{ij}q_j)$ is the only divergence that yields a closed-form Gibbs-type solution for the belief update and a consistent dual interpretation for the attention weights. Agents locally minimize their agent-specific variational free energy and the system of agents minimize their collective global free energies. This local-global coordination gives rise to the expected forward KL attention term.

In what follows we restrict to our “matched-fiber” framework, where all agents share the same fiber dimension K and representation ρ , so that gauge transport acts within a single vector space. The full generalization follows by applying the appropriate bundle morphisms/intertwiners.

The uniqueness of the term is conditional and follows from two assumptions: (i) $\mathcal{D}$ is local in c , of f -divergence form $\int q_i(c) f\left(\frac{q_i(c)}{\Omega_{ij}(c)q_j(c)}\right) dc$; and (ii) the coupling is linear in β_{ij} . Exponential-family closure of the resulting q_i^* is a *consequence* of the theorem (Step 2 of the proof produces the log-linear form automatically once $f'(t) = \log t$), not an additional hypothesis.

H.0.1 THE COUPLED VARIATIONAL PROBLEM

Each agent i minimizes a local free-energy functional:

$$F_i[\beta_i] = \min_{q_i} \left\{ D_{\text{KL}}(q_i \| p_i) + \sum_{j \neq i} \beta_{ij} \mathcal{D}(q_i, q_j) \right\}, \quad (68)$$

For convenience, we can decompose the KL as:

$$D_{\text{KL}}(q_i \| p_i) = \langle H_i \rangle_{q_i} + S(q_i) + \text{const}, \quad (69)$$

where

$$H_i(c) := -\log p_i(c) \quad (\text{local "energy"}), \quad (70)$$

$$S(q_i) := -\int q_i(c) \log q_i(c) dc \quad (\text{Shannon entropy}), \quad (71)$$

$$\langle H_i \rangle_{q_i} := \int q_i(c) H_i(c) dc \quad (\text{expected energy}). \quad (72)$$

This yields $D_{\text{KL}}(q_i \| p_i) = \int q_i(c) [\log q_i(c) - \log p_i(c)] dc$.

The attention weights $\beta_{ij} \geq 0$ (with $\sum_j \beta_{ij} = 1$) then are chosen by optimizing

$$\mathcal{J}_i(\beta_i) = \sum_{j \neq i} \beta_{ij} C_{ij} + \tau \sum_{j \neq i} \beta_{ij} \log \beta_{ij}, \quad (73)$$

where C_{ij} denotes the marginal cost of attending to agent j as described in the main paper.

Later we will identify this as

$$C_{ij} := \frac{\partial S_i}{\partial \beta_{ij}}, \quad (74)$$

such that attention weights allocate in proportion to marginal coordination penalties.

H.0.2 THE FORWARD KL AND THE GEOMETRIC-MEAN SOLUTION

Let $\mathcal{D}(q_i, q_j) = D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$, where

$$\frac{\delta D_{\text{KL}}(q_i \| \Omega_{ij} q_j)}{\delta q_i(c)} = \log \frac{q_i(c)}{\Omega_{ij}(c) q_j(c)} + 1.$$

The stationarity condition for (68) is

$$H_i(c) + \log q_i(c) + \sum_j \beta_{ij} \left[\log \frac{q_i(c)}{\Omega_{ij}(c) q_j(c)} + 1 \right] = \lambda_i, \quad (75)$$

with λ_i enforcing normalization.

Solving for $q_i(c)$ then yields a Boltzmann distribution whose mean field is the geometric average of the transported neighbor beliefs.

$$q_i^*(c) = \frac{1}{Z_i} e^{-H_i(c)/2} \prod_j [\Omega_{ij}(c) q_j(c)]^{\beta_{ij}/2}, \quad (76)$$

This structure is preserved only for the forward KL divergence; alternative divergences destroy the log-linearity of the exponent as previously discussed.

H.0.3 DUAL RELATION VIA THE ENVELOPE THEOREM

At the stationary value q_i^* , the envelope theorem implies

$$\frac{\partial F_i}{\partial \beta_{ij}} = \mathcal{D}(q_i^*, q_j) = D_{\text{KL}}(q_i^* \parallel \Omega_{ij} q_j), \quad (77)$$

such that the cost of increasing attention to j equals the forward KL divergence between the updated belief q_i^* and the transported neighbor $\Omega_{ij} q_j$.

This then identifies the attention cost with the KL.

H.0.4 CONDITIONAL UNIQUENESS THEOREM

Theorem. *Let $\mathcal{D}(q, p)$ be a convex f -divergence (Csiszár, 1967) of the form*

$$\mathcal{D}(q, p) = \int p(c) f\left(\frac{q(c)}{p(c)}\right) dc,$$

with $f : (0, \infty) \rightarrow \mathbb{R}$ convex, $f(1) = 0$, real-analytic on $(0, \infty)$, and, without loss of generality¹, $f'(1) = 0$. Suppose $\mathcal{D}$ is linear inside the free energy (68) and that the attention weights satisfy $\sum_j \beta_{ij} = 1$. Then the stationary solution q_i^* assumes the geometric-mean Boltzmann form

$$q_i^*(c) \propto p_i(c)^{1/2} \prod_j [\Omega_{ij}(c) q_j(c)]^{\beta_{ij}/2} \quad (78)$$

for all choices of prior p_i and neighbor beliefs $\{q_j\}$, if, and only if, $\mathcal{D}$ is the forward Kullback-Leibler divergence (up to the affine equivalence above):

$$\mathcal{D}(q, p) = D_{\text{KL}}(q \parallel p).$$

Proof. Step 1. The variational derivative of a general f -divergence.

With $\mathcal{D}(q_i, \Omega_{ij} q_j) = \int (\Omega_{ij} q_j) f(q_i / (\Omega_{ij} q_j)) dc$ and ratio $r(c) = q_i(c) / [\Omega_{ij}(c) q_j(c)]$, the functional derivative is

$$\frac{\delta \mathcal{D}}{\delta q_i(c)} = f'(r(c)). \quad (79)$$

The stationarity condition $\delta F_i / \delta q_i(c) = \lambda$ (with Lagrange multiplier λ) is then

$$\log \frac{q_i(c)}{p_i(c)} + 1 + \sum_j \beta_{ij} f'\left(\frac{q_i(c)}{\Omega_{ij}(c) q_j(c)}\right) = \lambda. \quad (80)$$

Step 2. Forward implication ($D_{\text{KL}} \Rightarrow$ geometric mean).

Next, for $\mathcal{D} = D_{\text{KL}}$, the generator is $f(t) = t \log t - t + 1$, yielding $f'(t) = \log t$. Substituting into (80) we have:

$$\log q_i - \log p_i + 1 + \sum_j \beta_{ij} (\log q_i - \log \Omega_{ij} q_j) = \lambda.$$

1. Any f -divergence is invariant under $f(t) \mapsto f(t) + c(t - 1)$ when q and p are normalized, so we may always choose the representative with $f'(1) = 0$.

However, since $\sum_j \beta_{ij} = 1$, the $\log q_i$ terms become $2 \log q_i$:

$$2 \log q_i = \log p_i + \sum_j \beta_{ij} \log(\Omega_{ij} q_j) + (\lambda - 1).$$

By exponentiating we find the geometric-mean form (78).

Step 3. Reverse implication (geometric mean $\Rightarrow D_{\text{KL}}$).

Assume the minimizer has the geometric mean form (78) for all priors p_i and neighbor beliefs $\{q_j\}$.

Taking logarithms yields:

$$2 \log q_i^*(c) = \log p_i(c) + \sum_j \beta_{ij} \log(\Omega_{ij}(c) q_j(c)) + C. \quad (81)$$

Next, rearranging and using $\sum_j \beta_{ij} = 1$ we have:

$$\log \frac{q_i^*}{p_i} + \sum_j \beta_{ij} \log \frac{q_i^*}{\Omega_{ij} q_j} = C'. \quad (82)$$

Subtracting (82) from the general stationarity condition (80) gives

$$\sum_j \beta_{ij} \left[f' \left(\frac{q_i^*(c)}{\Omega_{ij}(c) q_j(c)} \right) - \log \frac{q_i^*(c)}{\Omega_{ij}(c) q_j(c)} \right] = k, \quad (83)$$

with $k = \lambda - 1 - C'$ a constant independent of c . To isolate a single f' evaluation we exploit the freedom to choose the configuration: pick transported neighbors so that all $\Omega_{ij}(c) q_j(c)$ coincide pointwise with a common reference density $\bar{q}(c)$ — for example by setting $q_j(c) = \Omega_{ij}^{-1}(c) \bar{q}(c)$ for each j in the family of admissible neighbor beliefs — and let p_i vary so that $q_i^*(c)$ sweeps a range of values. All ratios $r_{ij}(c) = q_i^*(c)/\bar{q}(c)$ then coincide for fixed c , and (83) collapses via $\sum_j \beta_{ij} = 1$ to

$$f'(r(c)) - \log r(c) = k, \quad (84)$$

which holds for every value of r realized as (p_i, c) vary over their admissible ranges. The set of such ratios contains an open subinterval of $(0, \infty)$; by the real-analyticity hypothesis on f , the identity $f'(t) = \log t + k$ extends from this subinterval to all of $(0, \infty)$. Hence $f'(t) = \log t + k$ globally.

Next, by integrating we have:

$$f(t) = t \log t - t + kt + C_0.$$

If we impose $f'(1) = 0$ then $k = 0$. Similarly by imposing $f(1) = 0$ then $C_0 = 1$. Therefore, we have

$$f(t) = t \log t - t + 1,$$

which is the generator of $D_{\text{KL}}(q\|p)$. □

H.0.5 DUAL COST IDENTIFICATION

Having established D_{KL} is the unique divergence, the envelope theorem applied to the β_{ij} optimization confirms that the marginal attention cost is

$$C_{ij} \equiv \left. \frac{\partial F_i}{\partial \beta_{ij}} \right|_{q_i^*} = D_{\text{KL}}(q_i^* \parallel \Omega_{ij} q_j), \quad (85)$$

identifying the cost of attention with the information divergence between the agent’s updated belief and the neighbor’s transported signal.

H.0.6 INTERPRETATIONS

From the perspective of gauge invariance, the comparison is made between q_i and the transported $\Omega_{ij} q_j$ within the same frame (and similarly for $j \rightarrow i$); gauge covariance fixes what is compared, while variational duality fixes how it is compared. From the perspective of this duality, the forward KL is the only divergence that simultaneously yields a closed-form Boltzmann solution for q_i and a consistent dual cost $C_{ij} = \partial F_i / \partial \beta_{ij}$. From the perspective of information geometry, the forward KL is the Bregman divergence generated by the negative entropy potential, whose Hessian induces the Fisher-Rao metric and yields the mixed/exponential family-projection Pythagorean theorem. These global properties are not shared by generic f-divergences.

H.0.7 SUMMARY

Under the natural assumptions of locality, linear coupling, and exponential-family closure, the forward KL divergence

$$C_{ij} = D_{\text{KL}}(q_i \parallel \Omega_{ij} q_j)$$

is a necessary consequence of the variational and geometric structure underlying agent-agent coordination. By the $\text{GL}(K)$ invariance theorem (Theorem 1 in the main paper), the KL divergence is invariant under the full general linear group $\text{GL}(K)$, which includes compact subgroups such as $\text{SO}(N)$ and $\text{SU}(N)$ and more.

Appendix I. Derivation of the Prior-Bank Decode from Canonical Primitives

The prior-bank decode head computes per-token logits as

$$\ell_{i,v} = -\frac{D_{\text{KL}}(q_i^* \parallel \pi_v)}{\tau}, \quad v \in \{1, \dots, V\}, \quad (86)$$

where $q_i^* = \mathcal{N}(\mu_{q,i}, \Sigma_{q,i})$ is the converged variational posterior over the latent state at position i , $\pi_v = \mathcal{N}(\mu_{p,v}, \Sigma_{p,v})$ is the per-token prior carried in the prior bank, and τ is the decode softmax temperature. Equation (86) is the form used in the main paper (§??) and stated in the project’s architectural constraints. This appendix derives it from two canonical primitives: the Bayes-optimal Gaussian generative classifier (Bishop, 2006b, §4.2), and the accuracy-plus-complexity decomposition of the variational free energy Friston (2010); Blei

et al. (2017b). The derivation makes explicit which steps are textbook and which are the variational extension that distinguishes the prior-bank decode from a point-classifier readout.

I.1 Gaussian Generative Classifier in Bishop §4.2

For a point observation $x \in \mathbb{R}^K$ and a finite set of classes $\{C_v\}_{v=1}^V$ with Gaussian class-conditional densities $p(x | C_v) = \mathcal{N}(x; \mu_{p,v}, \Sigma_{p,v})$ and class prior $p(C_v) = \omega_v$, the Bayes posterior is

$$\log p(C_v | x) = \log p(x | C_v) + \log \omega_v - \log \sum_{v'} p(x | C_{v'}) \omega_{v'}, \quad (87)$$

and the unnormalized log-posterior reduces to

$$\log p(C_v | x) \propto_v -\frac{1}{2}(x - \mu_{p,v})^\top \Sigma_{p,v}^{-1}(x - \mu_{p,v}) - \frac{1}{2} \log |\Sigma_{p,v}| + \log \omega_v, \quad (88)$$

modulo additive terms that do not depend on v (the normalization constant and $-\frac{K}{2} \log 2\pi$). Equation (88) is the standard quadratic discriminant of (Bishop, 2006b, §4.2.1), recovering linear discriminant analysis under the tied-covariance specialization $\Sigma_{p,v} \equiv \Sigma_p$. The framework uses an isotropic class prior $\omega_v = 1/V$ so that $\log \omega_v$ is itself v -invariant and may be absorbed into the dropped constant.

I.2 Replacing the Point Observation with the Recognition Distribution

In the variational setting the model has access not to a point observation x but to a recognition distribution q_i^* over the latent. The canonical substitution is to score class v by the expected log-likelihood under q_i^* rather than by the log-likelihood at a single point,

$$\log p(C_v | q_i^*) \propto_v \mathbb{E}_{q_i^*}[\log p(x | C_v)]. \quad (89)$$

This substitution is the standard variational plug-in: x is no longer observed exactly, q_i^* is the inferential summary of what the system knows about it, and the predictive likelihood collapses to the expectation under that summary. The same construction underlies the predictive distribution in variational Bayes (Blei et al., 2017b, Eq. 5) and the expected free energy decomposition in active inference.

For Gaussian class conditionals the expectation in (89) is closed form. Writing

$$\log p(x | C_v) = -\frac{1}{2}(x - \mu_{p,v})^\top \Sigma_{p,v}^{-1}(x - \mu_{p,v}) - \frac{1}{2} \log |\Sigma_{p,v}| - \frac{K}{2} \log 2\pi,$$

and taking $\mathbb{E}_{q_i^*}[\cdot]$ term by term yields

$$\begin{aligned} \mathbb{E}_{q_i^*}[\log p(x | C_v)] &= -\frac{1}{2} \text{tr}(\Sigma_{p,v}^{-1} \Sigma_{q,i}) - \frac{1}{2} (\mu_{q,i} - \mu_{p,v})^\top \Sigma_{p,v}^{-1} (\mu_{q,i} - \mu_{p,v}) \\ &\quad - \frac{1}{2} \log |\Sigma_{p,v}| - \frac{K}{2} \log 2\pi, \end{aligned} \quad (90)$$

where the trace term arises from $\mathbb{E}_q[(x - \mu_p)^\top A(x - \mu_p)] = \text{tr}(A \Sigma_q) + (\mu_q - \mu_p)^\top A(\mu_q - \mu_p)$ with $A = \Sigma_{p,v}^{-1}$.

I.3 Identification with $-D_{\text{KL}}(q_i^* \parallel \pi_v)$

The closed form for the Gaussian KL divergence (Bishop, 2006b, App. B) is

$$D_{\text{KL}}(q_i^* \parallel \pi_v) = \frac{1}{2} \left[\text{tr}(\Sigma_{p,v}^{-1} \Sigma_{q,i}) + (\mu_{q,i} - \mu_{p,v})^\top \Sigma_{p,v}^{-1} (\mu_{q,i} - \mu_{p,v}) - K + \log \frac{|\Sigma_{p,v}|}{|\Sigma_{q,i}|} \right]. \quad (91)$$

Substituting (91) into (90) gives the identity

$$\mathbb{E}_{q_i^*}[\log p(x \mid C_v)] = -D_{\text{KL}}(q_i^* \parallel \pi_v) - H(q_i^*) + C, \quad (92)$$

where $H(q_i^*) = \frac{1}{2} \log((2\pi e)^K |\Sigma_{q,i}|)$ is the differential entropy of q_i^* and $C = -\frac{K}{2} \log 2\pi - \frac{K}{2}$ is a v -invariant scalar constant. Both $H(q_i^*)$ and C depend on the row index i but not on the column index v . Under the softmax over v ,

$$\frac{e^{\mathbb{E}_{q_i^*}[\log p(x \mid C_v)]/\tau}}{\sum_{v'} e^{\mathbb{E}_{q_i^*}[\log p(x \mid C_{v'})]/\tau}} = \frac{e^{-D_{\text{KL}}(q_i^* \parallel \pi_v)/\tau}}{\sum_{v'} e^{-D_{\text{KL}}(q_i^* \parallel \pi_{v'})/\tau}}, \quad (93)$$

the row-only contributions $-H(q_i^*) + C$ cancel between numerator and denominator. The resulting decoding distribution depends on i only through q_i^* , and the per-token logit (up to a v -invariant shift that the softmax absorbs) is

$$\ell_{i,v} = -\frac{D_{\text{KL}}(q_i^* \parallel \pi_v)}{\tau}, \quad (94)$$

which is the formula (86) in the main text.

I.4 Identification with the Accuracy Term of $\mathcal{F}$

The same expression admits a second canonical reading, via the accuracy-plus-complexity decomposition of the variational free energy Friston (2010); Blei et al. (2017b):

$$\mathcal{F}[q, o] = \mathbb{E}_q[-\log p(o \mid s)] + D_{\text{KL}}(q(s) \parallel p(s)), \quad (95)$$

where the first summand is the expected inaccuracy under the recognition distribution and the second is the complexity of the deviation from the prior. With observation space identified with the vocabulary, $o = v \in \{1, \dots, V\}$, and the per-token observation likelihood specified as a Gaussian template on the latent state,

$$p(o = v \mid s) = \frac{\mathcal{N}(s; \mu_{p,v}, \Sigma_{p,v})}{\sum_{v'} \mathcal{N}(s; \mu_{p,v'}, \Sigma_{p,v'})}, \quad (96)$$

the expected inaccuracy summand becomes

$$\mathbb{E}_{q_i^*}[-\log p(o = v \mid s)] = -\mathbb{E}_{q_i^*}[\log \mathcal{N}(s; \mu_{p,v}, \Sigma_{p,v})] + \mathbb{E}_{q_i^*}[\log Z(s)], \quad (97)$$

with $Z(s) = \sum_{v'} \mathcal{N}(s; \mu_{p,v'}, \Sigma_{p,v'})$. The first term is the negative of (90) and equals $D_{\text{KL}}(q_i^* \parallel \pi_v) + H(q_i^*) - C$ by (92); the second is v -invariant. The decode logit (94) is therefore the v -dependent portion of the accuracy summand of $\mathcal{F}$ under the per-token Gaussian observation model.

I.5 What this Derivation Establishes and What it Does Not

The derivation establishes that (86) is the composition of two canonical pieces: the Bishop quadratic discriminant for Gaussian class-conditionals (Bishop, 2006b, §4.2.1), and the standard variational plug-in $\mathbb{E}_q[\cdot]$ for point observations Blei et al. (2017b). The intermediate quantity $\mathbb{E}_{q_i^*}[\log p(x \mid C_v)]$ is in textbook form; the identity to $-D_{\text{KL}}(q_i^* \parallel \pi_v)$ up to row-only constants follows from the closed-form Gaussian KL (Bishop, 2006b, App. B); the equivalence under softmax follows from the cancellation in (93). Under the Gaussian token likelihood (96) the same logit is the v -dependent portion of the accuracy term of $\mathcal{F}$ Friston (2010).

Three scopes are worth stating explicitly. First, the derivation justifies the form of the readout, not its training objective: the M-step in the prior-bank implementation minimizes cross-entropy against the per-position target token, not the full free energy $\mathcal{F}$ summed over i and v , so the variational interpretation is structural (the readout is the accuracy term) rather than dynamical (training does not descend $\mathcal{F}$ directly). Second, the derivation is exact only when q_i^* and the $\{\pi_v\}$ lie on the same statistical submanifold; under the diagonal-covariance specialization the diagonal Gaussian submanifold is closed under the operations involved, but the full-covariance branch’s diagonal projection at decode is an approximation rather than a strict instance of the identity. Third, the temperature τ in (86) is parameterized in the implementation as $\tau = \tau_{\text{cfg}} \cdot e^{-s}$ with a learnable scalar s , mirroring the learnable attention temperature $\tau_{\text{attn}} = \kappa\sqrt{K}$ of the main paper; this learnable component is a single scalar on top of the canonical scaling and does not alter the derivation, but it should be reported when the formula is quoted as a comparison baseline.

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